

Snakemake

https://bioinfmsc8.bio.ed.ac.uk/AY25_BPSM22.html



THE UNIVERSITY
of EDINBURGH

edinburgh
genomics.

Thanks to Edinburgh Genomics: Tim Booth (historically Hywel Dunn-Davies, Katie Emelionova and Mark Blaxter)!

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Exercises

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<https://snakemake.readthedocs.io/en/stable/>

"The snakemake workflow management system is a tool to create reproducible and scalable data analyses. Workflows are described via a human readable, Python-based language.

They can be seamlessly scaled to server, cluster, grid and cloud environments, without the need to modify the workflow definition.

Finally, snakemake workflows can entail a description of required software, which will be automatically deployed to any execution environment."

Based on ideas from UNIX make...

- ...originally created by Stuart Feldman in 1976 at Bell Labs, now a standard build tool
- Started by Johannes Köster during his PhD
- v1.0 release in April 2012
- v5.4 release in March 2019
- currently **9.13.7**
- Works on Linux, Mac, Windows
- Free software...!

🏠 Let's install it!

Installation guide

Let's just try running it, most systems will probably have it, surely?!

```
# Logged into msc8
snakemake
Traceback (most recent call last):
  File "/localdisk/home/aivens2/.local/bin/snakemake", line 5, in <module>
    from snakemake import main
ModuleNotFoundError: No module named 'snakemake'
```

Hmm, looks like I used to have it, but there's no module there any more??

```
# Let's just install it...!
pip install --user snakemake
error: externally-managed-environment

× This environment is externally managed
  ↳ To install Python packages system-wide, try apt install
    python3-xyz, where xyz is the package you are trying to
    install.

    If you wish to install a non-Debian-packaged Python package,
    create a virtual environment using python3 -m venv path/to/venv.
    Then use path/to/venv/bin/python and path/to/venv/bin/pip. Make
    sure you have python3-full installed.

    If you wish to install a non-Debian packaged Python application,
    it may be easiest to use pipx install xyz, which will manage a
    virtual environment for you. Make sure you have pipx installed.

    See /usr/share/doc/python3.12/README.venv for more information.

note: If you believe this is a mistake, please contact your Python installation or OS dist
hint: See PEP 668 for the detailed specification.
```

MEH! Forgot about that!

Conda to the rescue, perhaps!?

```
# Is it installed on msc7 and msc8?
which conda
/localdisk/home/ubuntu-software/Anaconda3/bin/conda
```

```
conda --version
conda 25.7.0
```

```
# Where am I keeping things?
```

```
cat ${HOME}/.condarc
```

```
channels:  
  - conda-forge  
  - bioconda  
  - defaults
```

Curious, it doesn't say where my envs should be kept, but obviously, I have specified the channels in the past!

```
# What environments do I currently have?
```

```
conda env list
```

```
# conda environments:  
#  
NGG3                /home/aivens2/.conda/envs/NGG3  
fegenie             /home/aivens2/.conda/envs/fegenie  
foldseq             /home/aivens2/.conda/envs/foldseq  
base                /localdisk/home/ubuntu-software/Anaconda3  
qiime2-amplicon-2024.2 /localdisk/home/ubuntu-software/Anaconda3/envs/qiime2-amplicon-20
```

OK, looks like I am probably storing them locally, in my home space area.

```
# Let's remove one that I don't think I need any more....
```

```
conda env remove -y --name fegenie
```

```
CondaToSNonInteractiveError: Terms of Service have not been accepted for the following cha  
Please accept or remove them before proceeding:
```

- <https://repo.anaconda.com/pkg/main>
- <https://repo.anaconda.com/pkg/r>

```
To accept these channels' Terms of Service, run the following commands:  
conda tos accept --override-channels --channel https://repo.anaconda.com/pkg/main  
conda tos accept --override-channels --channel https://repo.anaconda.com/pkg/r
```

```
For information on safely removing channels from your conda configuration,  
please see the official documentation:
```

```
https://www.anaconda.com/docs/tools/working-with-conda/channels
```

```
# Do as they ask....
```

```
conda tos accept --override-channels --channel https://repo.anaconda.com/pkg/main
```

```
accepted Terms of Service for https://repo.anaconda.com/pkg/main
```

```
conda tos accept --override-channels --channel https://repo.anaconda.com/pkg/r
```

```
accepted Terms of Service for https://repo.anaconda.com/pkg/r
```

```
# Try again...
```

```
conda env remove -y --name fegenie
```

```
2 channel Terms of Service accepted
```

```
Remove all packages in environment /home/aivens2/.conda/envs/fegenie:
```

```
## Package Plan ##
```

```
environment location: /home/aivens2/.conda/envs/fegenie
```

```
The following packages will be REMOVED:
```

```
_libgcc_mutex-0.1-conda_forge
_openmp_mutex-4.5-2_gnu
_r-mutex-1.0.1-anaconda1
binutils_impl_linux-64-2.43-h4bf12b8_4
blast-2.16.0-h66d330f_5
bwidget-1.10.1-ha770c72_0
bzip2-1.0.8-h4bc722e_7
c-ares-1.34.5-hb9d3cd8_0
ca-certificates-2025.4.26-hbd8a1cb_0
cairo-1.16.0-ha61ee94_1012
curl-7.86.0-h2283fc2_1
diamond-2.1.11-h5ca1c30_2
entrez-direct-22.4-he881be0_0
fegenie-1.0-py39r40hdfd78af_0
font-ttf-dejavu-sans-mono-2.37-hab24e00_0
font-ttf-inconsolata-3.000-h77eed37_0
font-ttf-source-code-pro-2.038-h77eed37_0
font-ttf-ubuntu-0.83-h77eed37_3
fontconfig-2.15.0-h7e30c49_1
fonts-conda-ecosystem-1-0
fonts-conda-forge-1-0
freetype-2.13.3-ha770c72_1
fribidi-1.0.10-h36c2ea0_0
gcc_impl_linux-64-15.1.0-h4393ad2_2
gettext-0.24.1-h5888daf_0
gettext-tools-0.24.1-h5888daf_0
gfortran_impl_linux-64-15.1.0-h3b9cdf2_2
graphite2-1.3.13-h59595ed_1003
gsl-2.7-he838d99_0
gxx_impl_linux-64-15.1.0-h6a1bac1_2
harfbuzz-6.0.0-h8e241bc_0
hmmmer-3.4-hb6cb901_4
icu-70.1-h27087fc_0
jpeg-9e-h0b41bf4_3
kernel-headers_linux-64-3.10.0-he073ed8_18
keyutils-1.6.1-h166bdaf_0
krb5-1.19.3-h08a2579_0
ld_impl_linux-64-2.43-h712a8e2_4
lerc-4.0.0-h0aef613_1
libasprintf-0.24.1-h8e693c7_0
libasprintf-devel-0.24.1-h8e693c7_0
libblas-3.9.0-31_h59b9bed_openblas
libcblas-3.9.0-31_he106b2a_openblas
libcurl-7.86.0-h2283fc2_1
libdeflate-1.14-h166bdaf_0
libedit-3.1.20250104-pl5321h7949ede_0
libev-4.33-hd590300_2
libexpat-2.7.0-h5888daf_0
libffi-3.4.6-h2dba641_1
libfreetype-2.13.3-ha770c72_1
libfreetype6-2.13.3-h48d6fc4_1
libgcc-15.1.0-h767d61c_2
libgcc-devel_linux-64-15.1.0-h4c094af_102
libgcc-ng-15.1.0-h69a702a_2
libgettextpo-0.24.1-h5888daf_0
libgettextpo-devel-0.24.1-h5888daf_0
libgfortran-15.1.0-h69a702a_2
libgfortran-ng-15.1.0-h69a702a_2
libgfortran5-15.1.0-hcea5267_2
libglib-2.78.1-hebfc3b9_0
libgomp-15.1.0-h767d61c_2
libiconv-1.18-h4ce23a2_1
libidn2-2.3.8-ha4ef2c3_0
liblapack-3.9.0-31_h7ac8fdf_openblas
liblzma-5.8.1-hb9d3cd8_1
liblzma-devel-5.8.1-hb9d3cd8_1
libmpdec-4.0.0-hb9d3cd8_0
libnghttp2-1.64.0-h161d5f1_0
libopenblas-0.3.29-pthreads_h94d23a6_0
```

libpng-1.6.47-h943b412_0
libsanitizier-15.1.0-h97b714f_2
libsqlite-3.49.2-hee588c1_0
libssh2-1.11.1-hcf80075_0
libstdcxx-15.1.0-h8f9b012_2
libstdcxx-devel_linux-64-15.1.0-h4c094af_102
libstdcxx-ng-15.1.0-h4852527_2
libtiff-4.4.0-h82bc61c_5
libunistring-0.9.10-h7f98852_0
libuuid-2.38.1-h0b41bf4_0
libwebp-base-1.5.0-h851e524_0
libxcb-1.13-h7f98852_1004
libxcrypt-4.4.36-hd590300_1
libxml2-2.10.3-hca2bb57_4
libzlib-1.3.1-hb9d3cd8_2
make-4.4.1-hb9d3cd8_2
metabat2-2.12.1-1
mpi-1.0-openmpi
ncbi-vdb-3.2.1-h9948957_0
ncurses-6.5-h2d0b736_3
openmpi-4.1.6-hc5af2df_101
openssl-3.5.0-h7b32b05_1
pandoc-2.19.2-h32600fe_2
pango-1.50.14-hd33c08f_0
pcre2-10.40-hc3806b6_0
perl-5.32.1-7_hd590300_perl5
perl-archive-tar-2.40-pl5321hdfd78af_0
perl-carp-1.50-pl5321hd8ed1ab_0
perl-common-sense-3.75-pl5321hd8ed1ab_0
perl-compress-raw-bzip2-2.201-pl5321hbf60520_1
perl-compress-raw-zlib-2.202-pl5321hadc24fc_0
perl-encode-3.21-pl5321hb9d3cd8_1
perl-exporter-5.74-pl5321hd8ed1ab_0
perl-exporter-tiny-1.002002-pl5321hd8ed1ab_0
perl-extutils-makemaker-7.70-pl5321hd8ed1ab_0
perl-io-compress-2.201-pl5321h503566f_5
perl-io-zlib-1.14-pl5321hdfd78af_0
perl-json-4.10-pl5321hdfd78af_1
perl-json-xs-4.03-pl5321h9948957_4
perl-list-moreutils-0.430-pl5321hdfd78af_0
perl-list-moreutils-xs-0.430-pl5321h7b50bb2_5
perl-parent-0.243-pl5321hd8ed1ab_0
perl-pathtools-3.75-pl5321hb9d3cd8_2
perl-scalar-list-utils-1.69-pl5321hb9d3cd8_0
perl-storable-3.15-pl5321hb9d3cd8_2
perl-types-serialiser-1.01-pl5321hdfd78af_0
pip-25.1.1-pyh145f28c_0
pixman-0.46.0-h29eaf8c_0
prodigal-2.6.3-h7b50bb2_10
pthread-stubs-0.4-hb9d3cd8_1002
python-3.13.3-hf636f53_101_cp313
python_abi-3.13-7_cp313
r-argparse-2.1.6-r40hc72bb7e_0
r-askpass-1.1-r40hcfec24a_2
r-assertthat-0.2.1-r40hc72bb7e_2
r-backports-1.4.1-r40hcfec24a_0
r-base-4.0.5-hb87df5d_8
r-base64enc-0.1_3-r40hcfec24a_1004
r-bit-4.0.4-r40hcfec24a_0
r-bit64-4.0.5-r40hcfec24a_0
r-blob-1.2.3-r40hc72bb7e_0
r-brio-1.1.3-r40hcfec24a_0
r-broom-1.0.1-r40hc72bb7e_0
r-bslib-0.4.0-r40hc72bb7e_0
r-cachem-1.0.6-r40hcfec24a_0
r-callr-3.7.2-r40hc72bb7e_0
r-cellranger-1.1.0-r40hc72bb7e_1004
r-cli-3.4.1-r40h7525677_0
r-clipr-0.8.0-r40hc72bb7e_0
r-colorspace-2.0_3-r40h06615bd_0

r-cowplot-1.1.1-r40hc72bb7e_0
r-cpp11-0.4.2-r40hc72bb7e_0
r-crayon-1.5.1-r40hc72bb7e_0
r-curl-4.3.2-r40hcfec24a_0
r-data.table-1.14.2-r40hcfec24a_0
r-dbi-1.1.3-r40hc72bb7e_0
r-dbp1yr-2.2.1-r40hc72bb7e_0
r-desc-1.4.2-r40hc72bb7e_0
r-diffobj-0.3.5-r40hcfec24a_0
r-digest-0.6.29-r40h03ef668_0
r-dplyr-1.0.10-r40h7525677_0
r-dtp1yr-1.2.2-r40hc72bb7e_0
r-ellipsis-0.3.2-r40hcfec24a_0
r-evaluate-0.16-r40hc72bb7e_0
r-fansi-1.0.3-r40h06615bd_0
r-farver-2.1.1-r40h7525677_0
r-fastmap-1.1.0-r40h03ef668_0
r-findpython-1.0.7-r40hc72bb7e_0
r-forcats-0.5.2-r40hc72bb7e_0
r-fs-1.5.2-r40h7525677_1
r-gargle-1.2.1-r40hc72bb7e_0
r-generics-0.1.3-r40hc72bb7e_0
r-gg dendro-0.1.23-r40hc72bb7e_0
r-ggplot2-3.3.6-r40hc72bb7e_0
r-ggpubr-0.2.5-r40h6115d3f_1
r-ggrepel-0.9.1-r40h03ef668_0
r-ggsci-2.9-r40hc72bb7e_1003
r-ggsignif-0.6.3-r40hc72bb7e_0
r-glue-1.6.2-r40h06615bd_0
r-googledrive-2.0.0-r40hc72bb7e_0
r-goolesheets4-1.0.1-r40h785f33e_0
r-gridbase-0.4_7-r40hc72bb7e_1003
r-gridextra-2.3-r40hc72bb7e_1003
r-gtable-0.3.1-r40hc72bb7e_0
r-haven-2.5.0-r40h7525677_0
r-highr-0.9-r40hc72bb7e_0
r-hms-1.1.2-r40hc72bb7e_0
r-htmltools-0.5.3-r40h7525677_0
r-http-1.4.4-r40hc72bb7e_0
r-ids-1.0.1-r40hc72bb7e_1
r-isoband-0.2.5-r40h03ef668_0
r-jquerylib-0.1.4-r40hc72bb7e_0
r-jsonlite-1.8.0-r40h06615bd_0
r-knitr-1.40-r40hc72bb7e_0
r-labeling-0.4.2-r40hc72bb7e_1
r-lattice-0.20_45-r40hcfec24a_0
r-lifecycle-1.0.2-r40hc72bb7e_0
r-lubridate-1.8.0-r40h03ef668_0
r-magrittr-2.0.3-r40h06615bd_0
r-mass-7.3_58.1-r40h06615bd_0
r-matrix-1.4_1-r40h0154571_0
r-memoise-2.0.1-r40hc72bb7e_0
r-mgcv-1.8_40-r40h0154571_0
r-mime-0.12-r40hcfec24a_0
r-modelr-0.1.9-r40hc72bb7e_0
r-munsell-0.5.0-r40hc72bb7e_1004
r-nlme-3.1_159-r40h8da6f51_0
r-openssl-2.0.3-r40h1f3e0c5_0
r-pillar-1.8.1-r40hc72bb7e_0
r-pkgconfig-2.0.3-r40hc72bb7e_1
r-pkgload-1.3.0-r40hc72bb7e_0
r-plyr-1.8.7-r40h7525677_0
r-polynom-1.4_1-r40hc72bb7e_0
r-praise-1.0.0-r40hc72bb7e_1005
r-prettyunits-1.1.1-r40hc72bb7e_1
r-processx-3.7.0-r40h06615bd_0
r-progress-1.2.2-r40hc72bb7e_2
r-ps-1.7.1-r40h06615bd_0
r-purrr-0.3.4-r40hcfec24a_1
r-pvclust-2.2_0-r40ha770c72_1


```
r-r6-2.5.1-r40hc72bb7e_0
r-rappdirs-0.3.3-r40hcfec24a_0
r-rcolorbrewer-1.1_3-r40h785f33e_0
r-rcpp-1.0.9-r40h7525677_1
r-readr-2.1.2-r40h03ef668_0
r-readxl-1.4.1-r40hf23e330_0
r-rematch-1.0.1-r40hc72bb7e_1004
r-rematch2-2.1.2-r40hc72bb7e_1
r-reprex-2.0.2-r40hc72bb7e_0
r-reshape-0.8.9-r40hc72bb7e_0
r-reshape2-1.4.4-r40h03ef668_1
r-rlang-1.0.6-r40h7525677_0
r-rmarkdown-2.16-r40hc72bb7e_0
r-rprojroot-2.0.3-r40hc72bb7e_0
r-rstudioapi-0.14-r40hc72bb7e_0
r-rvest-1.0.3-r40hc72bb7e_0
r-sass-0.4.2-r40h7525677_0
r-scales-1.2.1-r40hc72bb7e_0
r-selectr-0.4_2-r40hc72bb7e_1
r-stringi-1.7.8-r40h30a9eb7_0
r-stringr-1.4.1-r40hc72bb7e_0
r-sys-3.4-r40hcfec24a_0
r-testthat-3.1.4-r40h7525677_0
r-tibble-3.1.8-r40h06615bd_0
r-tidyr-1.2.1-r40h7525677_0
r-tidymodels-1.1.2-r40hc72bb7e_0
r-tidyverse-1.3.2-r40hc72bb7e_0
r-tinytex-0.42-r40hc72bb7e_0
r-tzdb-0.3.0-r40h7525677_0
r-utf8-1.2.2-r40hcfec24a_0
r-uuid-1.1_0-r40h06615bd_0
r-vctrs-0.4.1-r40h7525677_0
r-viridislite-0.4.1-r40hc72bb7e_0
r-vroom-1.5.7-r40h03ef668_0
r-waldo-0.4.0-r40hc72bb7e_0
r-withr-2.5.0-r40hc72bb7e_0
r-xfun-0.33-r40h7525677_0
r-xml2-1.3.3-r40h7525677_1
r-yaml-2.3.5-r40h06615bd_0
readline-8.2-h8c095d6_2
rpsbproc-0.5.0-hd6d6fdc_3
sed-4.9-h6688a6e_0
sysroot_linux-64-2.17-h0157908_18
tk-8.6.13-noxft_hd72426e_102
tktable-2.10-h8bc8fbc_6
tzdata-2025b-h78e105d_0
wget-1.21.4-hda4d442_0
xorg-kbproto-1.0.7-hb9d3cd8_1003
xorg-libice-1.0.10-h7f98852_0
xorg-libsm-1.2.3-hd9c2040_1000
xorg-libx11-1.8.4-h0b41bf4_0
xorg-libxau-1.0.12-hb9d3cd8_0
xorg-libxdmcp-1.1.5-hb9d3cd8_0
xorg-libxext-1.3.4-h0b41bf4_2
xorg-libxrender-0.9.10-h7f98852_1003
xorg-libxt-1.3.0-hd590300_0
xorg-renderproto-0.11.1-hb9d3cd8_1003
xorg-xextproto-7.3.0-hb9d3cd8_1004
xorg-xproto-7.0.31-hb9d3cd8_1008
xz-5.8.1-hbcc6ac9_1
xz-gpl-tools-5.8.1-hbcc6ac9_1
xz-tools-5.8.1-hb9d3cd8_1
zlib-1.3.1-hb9d3cd8_2
zstd-1.5.7-hb8e6e7a_2
```

Downloading and Extracting Packages:

Preparing transaction: done

```
Verifying transaction: done
Executing transaction: done
```

Actually, let's get rid of all of them (apart from base, obviously!), they are no longer needed...

Remove multiple environments at the same time with a simple loop!

```
for env_name in NGG3 foldseq qiime2-amplicon-2024.2
do
conda env remove -y --name ${env_name}
done
```

2 channel Terms of Service accepted

Remove all packages in environment /home/aivens2/.conda/envs/NGG3:

Package Plan

environment location: /home/aivens2/.conda/envs/NGG3

The following packages will be REMOVED:

```
_libgcc_mutex-0.1-conda_forge
_openmp_mutex-4.5-1_gnu
arrow-cpp-3.0.0-py37hced866c_0
biopython-1.78-py37h7b6447c_0
blas-1.0-mkl
boost-cpp-1.73.0-h27cfd23_11
brotli-1.0.9-he6710b0_2
bzip2-1.0.8-h7f98852_4
c-ares-1.17.1-h36c2ea0_0
ca-certificates-2021.1.19-h06a4308_0
certifi-2020.12.5-py37h06a4308_0
curl-7.71.1-he644dc0_8
cyclcr-0.10.0-py37_0
dbus-1.13.18-hb2f20db_0
deprecated-1.2.11-pyhd3eb1b0_0
....
....
zeromq-4.3.5-h59595ed_0
zipp-3.17.0-pyhd8ed1ab_0
zlib-1.2.13-hd590300_5
zstandard-0.22.0-py38ha98ab4e_0
zstd-1.5.5-hfc55251_0
```

Downloading and Extracting Packages:

```
Preparing transaction: done
Verifying transaction: failed
```

EnvironmentNotWritableError: The current user does not have write permissions to the target environment location: /localdisk/home/ubuntu-software/Anaconda3/envs/qiime2-amplicon-2024.2
uid: 864524
gid: 864524

What happened?

```
conda env list
```

```
# conda environments:
#
```

```
base /localdisk/home/ubuntu-software/Anaconda3
qiime2-amplicon-2024.2 /localdisk/home/ubuntu-software/Anaconda3/envs/qiime2-amplicon-20
```

The `qiime` one isn't in my home space, and I don't have permission to remove it.

Anyway, OK, now onto what we actually wanted to do: run/install `snakemake`, but let's be a bit more structured....

```
# Make some folders....
```

```
mkdir ${HOME}/MyCondaEnvs/
```

```
mkdir ${HOME}/MyCondaPacks/
```

```
# Tell conda where we want our environments to be kept
```

```
conda config --add envs_dirs ${HOME}/MyCondaEnvs/
```

```
# Tell conda where we want our packages to be kept
```

```
conda config --add pkgs_dirs ${HOME}/MyCondaPacks/
```

```
# What channels do we have?
```

```
conda config --show channels
```

```
channels:
- conda-forge
- bioconda
- defaults
```

```
# What channel priorities do we have?
```

```
conda config --get channels
```

```
--add channels 'defaults'    # lowest priority
--add channels 'bioconda'
--add channels 'conda-forge'  # highest priority
```

```
# We believe bioconda should be highest priority!
```

```
conda config --add channels bioconda
```

```
Warning: 'bioconda' already in 'channels' list, moving to the top
```

```
# What channel priorities do we have now?
```

```
conda config --get channels
```

```
--add channels 'defaults'    # lowest priority
--add channels 'conda-forge'
--add channels 'bioconda'    # highest priority
```

```
# How else could we have done this?
```

```
cat ${HOME}/.condarc
```

```
channels:
- bioconda
- conda-forge
- defaults
envs_dirs:
- /home/aivens2/MyCondaEnvs/
```

```
pkgs_dirs:  
- /home/aivens2/MyCondaPacks
```

I think we are ready now...?

```
conda create --name snakemake snakemake  
  
2 channel Terms of Service accepted  
Retrieving notices: done  
Channels:  
- bioconda  
- conda-forge  
- defaults  
Platform: linux-64  
Collecting package metadata (repodata.json): done  
Solving environment: done  
  
==> WARNING: A newer version of conda exists. <==  
current version: 25.7.0  
latest version: 25.11.0  
  
Please update conda by running  
  
$ conda update -n base -c defaults conda  
  
## Package Plan ##  
  
environment location: /home/aivens2/MyCondaEnvs/snakemake  
  
added / updated specs:  
- snakemake  
  
The following packages will be downloaded:  
  
package | build | size | channel  
-----|-----|-----|-----  
_libgcc_mutex-0.1 | conda_forge | 3 KB | conda-forge  
_openmp_mutex-4.5 | 2_gnu | 23 KB | conda-forge  
amply-0.1.6 | pyhd8ed1ab_1 | 21 KB | conda-forge  
annotated-types-0.7.0 | pyhd8ed1ab_1 | 18 KB | conda-forge  
appdirs-1.4.4 | pyhd8ed1ab_1 | 14 KB | conda-forge  
argparse-dataclass-2.0.0 | pyhd8ed1ab_1 | 13 KB | conda-forge  
attrs-25.4.0 | pyh71513ae_0 | 59 KB | conda-forge  
brotli-python-1.2.0 | py313hf159716_1 | 359 KB | conda-forge  
bzip2-1.0.8 | hda65f42_8 | 254 KB | conda-forge  
ca-certificates-2025.11.12 | hbd8a1cb_0 | 149 KB | conda-forge  
certifi-2025.11.12 | pyhd8ed1ab_0 | 153 KB | conda-forge  
cffi-2.0.0 | py313hf46b229_1 | 291 KB | conda-forge  
charset-normalizer-3.4.4 | pyhd8ed1ab_0 | 50 KB | conda-forge  
click-8.3.1 | pyh707e725_0 | 90 KB | conda-forge  
coin-or-cbc-2.10.12 | h4d16d09_4 | 889 KB | conda-forge  
coin-or-cgl-0.60.9 | hc46dffc_6 | 521 KB | conda-forge  
coin-or-clp-1.17.10 | hc03379b_3 | 1.1 MB | conda-forge  
coin-or-osi-0.108.11 | hf4fecb4_8 | 368 KB | conda-forge  
coin-or-utis-2.11.12 | hc93afbd_7 | 649 KB | conda-forge  
colorama-0.4.6 | pyhd8ed1ab_1 | 26 KB | conda-forge  
coloredlogs-15.0.1 | pyhd8ed1ab_4 | 43 KB | conda-forge  
conda-inject-1.3.2 | pyhd8ed1ab_0 | 10 KB | conda-forge  
configargparse-1.7.1 | pyhe01879c_0 | 40 KB | conda-forge  
connection_pool-0.0.3 | pyhd3deb0d_0 | 8 KB | conda-forge  
docutils-0.22.3 | pyhd8ed1ab_0 | 427 KB | conda-forge  
dpath-2.2.0 | pyha770c72_1 | 21 KB | conda-forge  
eido-0.2.4 | pyhd8ed1ab_0 | 20 KB | conda-forge  
exceptiongroup-1.3.1 | pyhd8ed1ab_0 | 21 KB | conda-forge
```

gitdb-4.0.12		pyhd8ed1ab_0	52 KB	conda-forge
gitpython-3.1.45		pyhff2d567_0	154 KB	conda-forge
h2-4.3.0		pyhcf101f3_0	94 KB	conda-forge
hpack-4.1.0		pyhd8ed1ab_0	30 KB	conda-forge
humanfriendly-10.0		pyh707e725_8	72 KB	conda-forge
hyperframe-6.1.0		pyhd8ed1ab_0	17 KB	conda-forge
icu-75.1		he02047a_0	11.6 MB	conda-forge
idna-3.11		pyhd8ed1ab_0	50 KB	conda-forge
immutable-0.21		py313h07c4f96_2	53 KB	conda-forge
iniconfig-2.3.0		pyhd8ed1ab_0	13 KB	conda-forge
jinja2-3.1.6		pyhd8ed1ab_0	110 KB	conda-forge
jsonschema-4.25.1		pyhe01879c_0	80 KB	conda-forge
jsonschema-specifications-2025.9.1		pyhcf101f3_0	19 KB	conda-forge
jupyter_core-5.9.1		pyhc90fa1f_0	64 KB	conda-forge
ld_impl_linux-64-2.45		default_hbd61a6d_104	709 KB	conda-forge
libblas-3.11.0		2_h4a7cf45_openblas	18 KB	conda-forge
libcbblas-3.11.0		2_h0358290_openblas	18 KB	conda-forge
libexpat-2.7.3		hecca717_0	75 KB	conda-forge
libffi-3.5.2		h9ec8514_0	56 KB	conda-forge
libgcc-15.2.0		he0feb66_13	1016 KB	conda-forge
libgcc-ng-15.2.0		h69a702a_13	26 KB	conda-forge
libgfortran-15.2.0		h69a702a_13	26 KB	conda-forge
libgfortran5-15.2.0		h68bc16d_13	2.4 MB	conda-forge
libgomp-15.2.0		he0feb66_13	590 KB	conda-forge
liblapack-3.11.0		2_h47877c9_openblas	18 KB	conda-forge
liblapacke-3.11.0		2_h6ae95b6_openblas	18 KB	conda-forge
liblzma-5.8.1		hb9d3cd8_2	110 KB	conda-forge
libmpdec-4.0.0		hb9d3cd8_0	89 KB	conda-forge
libopenblas-0.3.30		pthread_h94d23a6_4	5.7 MB	conda-forge
libsqlite-3.51.0		hee844dc_0	923 KB	conda-forge
libstdcxx-15.2.0		h934c35e_13	5.6 MB	conda-forge
libstdcxx-ng-15.2.0		hdf11a46_13	26 KB	conda-forge
libuuid-2.41.2		he9a06e4_0	36 KB	conda-forge
libzlib-1.3.1		hb9d3cd8_2	60 KB	conda-forge
logmuse-0.2.8		pyhd8ed1ab_1	13 KB	conda-forge
markdown-it-py-4.0.0		pyhd8ed1ab_0	63 KB	conda-forge
markupsafe-3.0.3		py313h3dea7bd_0	25 KB	conda-forge
mdurl-0.1.2		pyhd8ed1ab_1	14 KB	conda-forge
nbformat-5.10.4		pyhd8ed1ab_1	99 KB	conda-forge
ncurses-6.5		h2d0b736_3	871 KB	conda-forge
numpy-2.3.5		py313hf6604e3_0	8.5 MB	conda-forge
openssl-3.6.0		h26f9b46_0	3.0 MB	conda-forge
packaging-25.0		pyh29332c3_1	61 KB	conda-forge
pandas-2.3.3		py313h08cd8bf_1	14.4 MB	conda-forge
pephubclient-0.4.4		pyhd8ed1ab_1	22 KB	conda-forge
peppy-0.40.8		pyhd8ed1ab_0	29 KB	conda-forge
pip-25.3		pyh145f28c_0	1.1 MB	conda-forge
plac-1.4.5		pyhd8ed1ab_0	26 KB	conda-forge
platformdirs-4.5.0		pyhcf101f3_0	23 KB	conda-forge
pluggy-1.6.0		pyhd8ed1ab_0	24 KB	conda-forge
psutil-7.1.3		py313h54dd161_0	490 KB	conda-forge
pulp-2.8.0		py313hf1034c9_3	223 KB	conda-forge
pyparser-2.22		pyh29332c3_1	108 KB	conda-forge
pydantic-2.12.5		pyhd8ed1ab_0	313 KB	conda-forge
pydantic-core-2.41.5		py313h843e2db_1	1.9 MB	conda-forge
pygments-2.19.2		pyhd8ed1ab_0	868 KB	conda-forge
pyparsing-3.2.5		pyhcf101f3_0	102 KB	conda-forge
pysocks-1.7.1		pyha55dd90_7	21 KB	conda-forge
pytest-9.0.1		pyhcf101f3_0	292 KB	conda-forge
python-3.13.9		hc97d973_101_cp313	35.5 MB	conda-forge
python-dateutil-2.9.0.post0		pyhe01879c_2	228 KB	conda-forge
python-fastjsonschema-2.21.2		pyhe01879c_0	239 KB	conda-forge
python-tzdata-2025.2		pyhd8ed1ab_0	141 KB	conda-forge
python_abi-3.13		8_cp313	7 KB	conda-forge
pytz-2025.2		pyhd8ed1ab_0	185 KB	conda-forge
pyyaml-6.0.3		py313h3dea7bd_0	202 KB	conda-forge
readline-8.2		h8c095d6_2	276 KB	conda-forge
referencing-0.37.0		pyhcf101f3_0	51 KB	conda-forge
requests-2.32.5		pyhd8ed1ab_0	58 KB	conda-forge
reretry-0.11.8		pyhd8ed1ab_1	12 KB	conda-forge

rich-14.2.0		pyhcf101f3_0	196 KB	conda-forge
rpds-py-0.29.0		py313h843e2db_0	376 KB	conda-forge
shellingham-1.5.4		pyhd8ed1ab_2	15 KB	conda-forge
six-1.17.0		pyhe01879c_1	18 KB	conda-forge
slack-sdk-3.39.0		pyhd8ed1ab_0	154 KB	conda-forge
slack_sdk-3.39.0		hd8ed1ab_0	7 KB	conda-forge
smart_open-7.5.0		pyhcf101f3_0	55 KB	conda-forge
s mmap-5.0.2		pyhd8ed1ab_0	25 KB	conda-forge
snakemake-9.13.7		hdfd78af_0	10 KB	bioconda
snakemake-interface-common-1.22.0		pyhd4c3c12_0	21 KB	bioconda
snakemake-interface-executor-plugins-9.3.9		pyhdfd78af_0	22 KB	bioconda
snakemake-interface-logger-plugins-2.0.0		pyhd4c3c12_0	18 KB	bioconda
snakemake-interface-report-plugins-1.3.0		pyhd4c3c12_0	14 KB	bioconda
snakemake-interface-scheduler-plugins-2.0.2		pyhd4c3c12_0	16 KB	bioconda
snakemake-interface-storage-plugins-4.2.3		pyhd4c3c12_0	21 KB	bioconda
snakemake-minimal-9.13.7		pyhdfd78af_0	846 KB	bioconda
tabulate-0.9.0		pyhd8ed1ab_2	37 KB	conda-forge
throttler-1.2.2		pyhd8ed1ab_0	12 KB	conda-forge
tk-8.6.13		noxft_ha0e22de_103	3.1 MB	conda-forge
tomli-2.3.0		pyhcf101f3_0	20 KB	conda-forge
traitlets-5.14.3		pyhd8ed1ab_1	107 KB	conda-forge
typer-0.20.0		pyhefaf540_1	78 KB	conda-forge
typer-slim-0.20.0		pyhcf101f3_1	47 KB	conda-forge
typer-slim-standard-0.20.0		h4daf872_1	5 KB	conda-forge
typing-extensions-4.15.0		h396c80c_0	89 KB	conda-forge
typing-inspection-0.4.2		pyhd8ed1ab_1	18 KB	conda-forge
typing_extensions-4.15.0		pyhcf101f3_0	50 KB	conda-forge
tzdata-2025b		h78e105d_0	120 KB	conda-forge
ubiquerg-0.8.0		pyhd8ed1ab_0	20 KB	conda-forge
urllib3-2.5.0		pyhd8ed1ab_0	99 KB	conda-forge
veracitools-0.1.3		py_0	6 KB	conda-forge
wrapt-1.17.3		py313h07c4f96_1	63 KB	conda-forge
yaml-0.2.5		h280c20c_3	83 KB	conda-forge
yte-1.8.1		pyha770c72_1	16 KB	conda-forge
zstandard-0.25.0		py313h54dd161_1	460 KB	conda-forge
zstd-1.5.7		hb8e6e7a_2	554 KB	conda-forge

Total:			111.3 MB	

The following NEW packages will be INSTALLED:

_libgcc_mutex	conda-forge/linux-64::_libgcc_mutex-0.1-conda_forge
_openmp_mutex	conda-forge/linux-64::_openmp_mutex-4.5-2_gnu
amply	conda-forge/noarch::amply-0.1.6-pyhd8ed1ab_1
annotated-types	conda-forge/noarch::annotated-types-0.7.0-pyhd8ed1ab_1
appdirs	conda-forge/noarch::appdirs-1.4.4-pyhd8ed1ab_1
argparse-dataclass	conda-forge/noarch::argparse-dataclass-2.0.0-pyhd8ed1ab_1
attrs	conda-forge/noarch::attrs-25.4.0-pyh71513ae_0
brotli-python	conda-forge/linux-64::brotli-python-1.2.0-py313hf159716_1
bzip2	conda-forge/linux-64::bzip2-1.0.8-hda65f42_8
ca-certificates	conda-forge/noarch::ca-certificates-2025.11.12-hbd8a1cb_0
certifi	conda-forge/noarch::certifi-2025.11.12-pyhd8ed1ab_0
cffi	conda-forge/linux-64::cffi-2.0.0-py313hf46b229_1
charset-normalizer	conda-forge/noarch::charset-normalizer-3.4.4-pyhd8ed1ab_0
click	conda-forge/noarch::click-8.3.1-pyh707e725_0
coin-or-cbc	conda-forge/linux-64::coin-or-cbc-2.10.12-h4d16d09_4
coin-or-cgl	conda-forge/linux-64::coin-or-cgl-0.60.9-hc46dffc_6
coin-or-clp	conda-forge/linux-64::coin-or-clp-1.17.10-hc03379b_3
coin-or-osi	conda-forge/linux-64::coin-or-osi-0.108.11-hf4fecb4_8
coin-or-utils	conda-forge/linux-64::coin-or-utils-2.11.12-hc93afbd_7
colorama	conda-forge/noarch::colorama-0.4.6-pyhd8ed1ab_1
coloredlogs	conda-forge/noarch::coloredlogs-15.0.1-pyhd8ed1ab_4
conda-inject	conda-forge/noarch::conda-inject-1.3.2-pyhd8ed1ab_0
configargparse	conda-forge/noarch::configargparse-1.7.1-pyhe01879c_0
connection_pool	conda-forge/noarch::connection_pool-0.0.3-pyhd3deb0d_0
docutils	conda-forge/noarch::docutils-0.22.3-pyhd8ed1ab_0
dpath	conda-forge/noarch::dpath-2.2.0-pyha770c72_1
eido	conda-forge/noarch::eido-0.2.4-pyhd8ed1ab_0
exceptiongroup	conda-forge/noarch::exceptiongroup-1.3.1-pyhd8ed1ab_0
gitdb	conda-forge/noarch::gitdb-4.0.12-pyhd8ed1ab_0

gitpython	conda-forge/noarch::gitpython-3.1.45-pyhff2d567_0
h2	conda-forge/noarch::h2-4.3.0-pyhcf101f3_0
hpack	conda-forge/noarch::hpack-4.1.0-pyhd8ed1ab_0
humanfriendly	conda-forge/noarch::humanfriendly-10.0-pyh707e725_8
hyperframe	conda-forge/noarch::hyperframe-6.1.0-pyhd8ed1ab_0
icu	conda-forge/linux-64::icu-75.1-he02047a_0
idna	conda-forge/noarch::idna-3.11-pyhd8ed1ab_0
immutable	conda-forge/linux-64::immutable-0.21-py313h07c4f96_2
iniconfig	conda-forge/noarch::iniconfig-2.3.0-pyhd8ed1ab_0
jinja2	conda-forge/noarch::jinja2-3.1.6-pyhd8ed1ab_0
jsonschema	conda-forge/noarch::jsonschema-4.25.1-pyhe01879c_0
jsonschema-specif~	conda-forge/noarch::jsonschema-specifications-2025.9.1-pyhcf101f3_0
jupyter_core	conda-forge/noarch::jupyter_core-5.9.1-pyhcf90fa1f_0
ld_impl_linux-64	conda-forge/linux-64::ld_impl_linux-64-2.45-default_hbd61a6d_104
libblas	conda-forge/linux-64::libblas-3.11.0-2_h4a7cf45_openblas
libcbblas	conda-forge/linux-64::libcbblas-3.11.0-2_h0358290_openblas
libexpat	conda-forge/linux-64::libexpat-2.7.3-hecca717_0
libffi	conda-forge/linux-64::libffi-3.5.2-h9ec8514_0
libgcc	conda-forge/linux-64::libgcc-15.2.0-he0feb66_13
libgcc-ng	conda-forge/linux-64::libgcc-ng-15.2.0-h69a702a_13
libgfortran	conda-forge/linux-64::libgfortran-15.2.0-h69a702a_13
libgfortran5	conda-forge/linux-64::libgfortran5-15.2.0-h68bc16d_13
libgomp	conda-forge/linux-64::libgomp-15.2.0-he0feb66_13
liblapack	conda-forge/linux-64::liblapack-3.11.0-2_h47877c9_openblas
liblapacke	conda-forge/linux-64::liblapacke-3.11.0-2_h6ae95b6_openblas
liblzma	conda-forge/linux-64::liblzma-5.8.1-hb9d3cd8_2
libmpdec	conda-forge/linux-64::libmpdec-4.0.0-hb9d3cd8_0
libopenblas	conda-forge/linux-64::libopenblas-0.3.30-pthreads_h94d23a6_4
libsqlite	conda-forge/linux-64::libsqlite-3.51.0-hee844dc_0
libstdcxx	conda-forge/linux-64::libstdcxx-15.2.0-h934c35e_13
libstdcxx-ng	conda-forge/linux-64::libstdcxx-ng-15.2.0-hdf11a46_13
libuuid	conda-forge/linux-64::libuuid-2.41.2-he9a06e4_0
libzlib	conda-forge/linux-64::libzlib-1.3.1-hb9d3cd8_2
logmuse	conda-forge/noarch::logmuse-0.2.8-pyhd8ed1ab_1
markdown-it-py	conda-forge/noarch::markdown-it-py-4.0.0-pyhd8ed1ab_0
markupsafe	conda-forge/linux-64::markupsafe-3.0.3-py313h3dea7bd_0
mdurl	conda-forge/noarch::mdurl-0.1.2-pyhd8ed1ab_1
nbformat	conda-forge/noarch::nbformat-5.10.4-pyhd8ed1ab_1
ncurses	conda-forge/linux-64::ncurses-6.5-h2d0b736_3
numpy	conda-forge/linux-64::numpy-2.3.5-py313hf6604e3_0
openssl	conda-forge/linux-64::openssl-3.6.0-h26f9b46_0
packaging	conda-forge/noarch::packaging-25.0-pyh29332c3_1
pandas	conda-forge/linux-64::pandas-2.3.3-py313h08cd8bf_1
pephubclient	conda-forge/noarch::pephubclient-0.4.4-pyhd8ed1ab_1
peppy	conda-forge/noarch::peppy-0.40.8-pyhd8ed1ab_0
pip	conda-forge/noarch::pip-25.3-pyh145f28c_0
plac	conda-forge/noarch::plac-1.4.5-pyhd8ed1ab_0
platformdirs	conda-forge/noarch::platformdirs-4.5.0-pyhcf101f3_0
pluggy	conda-forge/noarch::pluggy-1.6.0-pyhd8ed1ab_0
psutil	conda-forge/linux-64::psutil-7.1.3-py313h54dd161_0
pulp	conda-forge/linux-64::pulp-2.8.0-py313hf1034c9_3
pyparser	conda-forge/noarch::pyparser-2.22-pyh29332c3_1
pydantic	conda-forge/noarch::pydantic-2.12.5-pyhd8ed1ab_0
pydantic-core	conda-forge/linux-64::pydantic-core-2.41.5-py313h843e2db_1
pygments	conda-forge/noarch::pygments-2.19.2-pyhd8ed1ab_0
pyparsing	conda-forge/noarch::pyparsing-3.2.5-pyhcf101f3_0
pysocks	conda-forge/noarch::pysocks-1.7.1-pyha55dd90_7
pytest	conda-forge/noarch::pytest-9.0.1-pyhcf101f3_0
python	conda-forge/linux-64::python-3.13.9-hc97d973_101_cp313
python-dateutil	conda-forge/noarch::python-dateutil-2.9.0.post0-pyhe01879c_2
python-fastjsonsc~	conda-forge/noarch::python-fastjsonschema-2.21.2-pyhe01879c_0
python-tzdata	conda-forge/noarch::python-tzdata-2025.2-pyhd8ed1ab_0
python_abi	conda-forge/noarch::python_abi-3.13-8_cp313
pytz	conda-forge/noarch::pytz-2025.2-pyhd8ed1ab_0
pyyaml	conda-forge/linux-64::pyyaml-6.0.3-py313h3dea7bd_0
readline	conda-forge/linux-64::readline-8.2-h8c095d6_2
referencing	conda-forge/noarch::referencing-0.37.0-pyhcf101f3_0
requests	conda-forge/noarch::requests-2.32.5-pyhd8ed1ab_0
reretry	conda-forge/noarch::reretry-0.11.8-pyhd8ed1ab_1
rich	conda-forge/noarch::rich-14.2.0-pyhcf101f3_0


```
rpds-py                conda-forge/linux-64::rpds-py-0.29.0-py313h843e2db_0
shellingham            conda-forge/noarch::shellingham-1.5.4-pyhd8ed1ab_2
six                   conda-forge/noarch::six-1.17.0-pyhe01879c_1
slack-sdk             conda-forge/noarch::slack-sdk-3.39.0-pyhd8ed1ab_0
slack_sdk            conda-forge/noarch::slack_sdk-3.39.0-hd8ed1ab_0
smart_open            conda-forge/noarch::smart_open-7.5.0-pyhcf101f3_0
s mmap               conda-forge/noarch::s mmap-5.0.2-pyhd8ed1ab_0
snakemake             bioconda/noarch::snakemake-9.13.7-hdfd78af_0
snakemake-interfa~    bioconda/noarch::snakemake-interface-common-1.22.0-pyhd4c3c12_0
snakemake-interfa~    bioconda/noarch::snakemake-interface-executor-plugins-9.3.9-pyhd78af78a
snakemake-interfa~    bioconda/noarch::snakemake-interface-logger-plugins-2.0.0-pyhd4c3c12_
snakemake-interfa~    bioconda/noarch::snakemake-interface-report-plugins-1.3.0-pyhd4c3c12_
snakemake-interfa~    bioconda/noarch::snakemake-interface-scheduler-plugins-2.0.2-pyhd4c3c
snakemake-interfa~    bioconda/noarch::snakemake-interface-storage-plugins-4.2.3-pyhd4c3c12
snakemake-minimal     bioconda/noarch::snakemake-minimal-9.13.7-pyhd78af78af_0
tabulate             conda-forge/noarch::tabulate-0.9.0-pyhd8ed1ab_2
throttler            conda-forge/noarch::throttler-1.2.2-pyhd8ed1ab_0
tk                   conda-forge/linux-64::tk-8.6.13-noxft_ha0e22de_103
tomli                conda-forge/noarch::tomli-2.3.0-pyhcf101f3_0
traitlets            conda-forge/noarch::traitlets-5.14.3-pyhd8ed1ab_1
typer                conda-forge/noarch::typer-0.20.0-pyhef540_1
typer-slim           conda-forge/noarch::typer-slim-0.20.0-pyhcf101f3_1
typer-slim-standa~    conda-forge/noarch::typer-slim-standard-0.20.0-h4daf872_1
typing-extensions     conda-forge/noarch::typing-extensions-4.15.0-h396c80c_0
typing-inspection     conda-forge/noarch::typing-inspection-0.4.2-pyhd8ed1ab_1
typing_extensions     conda-forge/noarch::typing_extensions-4.15.0-pyhcf101f3_0
tzdata              conda-forge/noarch::tzdata-2025b-h78e105d_0
ubiquerg            conda-forge/noarch::ubiquerg-0.8.0-pyhd8ed1ab_0
urllib3             conda-forge/noarch::urllib3-2.5.0-pyhd8ed1ab_0
veracitools         conda-forge/noarch::veracitools-0.1.3-py_0
wrapt               conda-forge/linux-64::wrapt-1.17.3-py313h07c4f96_1
yaml                conda-forge/linux-64::yaml-0.2.5-h280c20c_3
yte                 conda-forge/noarch::yte-1.8.1-pyha770c72_1
zstandard           conda-forge/linux-64::zstandard-0.25.0-py313h54dd161_1
zstd                conda-forge/linux-64::zstd-1.5.7-hb8e6e7a_2
```

Proceed ([y]/n)?

Downloading and Extracting Packages:

```
python-3.13.9          | 35.5 MB | #####
pandas-2.3.3          | 14.4 MB | #####
icu-75.1              | 11.6 MB | #####
numpy-2.3.5           | 8.5 MB  | #####
libopenblas-0.3.30    | 5.7 MB  | #####
libstdcxx-15.2.0      | 5.6 MB  | #####
tk-8.6.13             | 3.1 MB  | #####
openssl-3.6.0         | 3.0 MB  | #####
libgfortran5-15.2.0   | 2.4 MB  | #####
.....
```

Downloading and Extracting Packages:

```
Preparing transaction: done
Verifying transaction: done
Executing transaction: done
#
# To activate this environment, use
#
#     $ conda activate snakemake
#
# To deactivate an active environment, use
#
#     $ conda deactivate
```

ls ~/MyCondaEnvs/

snakemake

Is ~/MyCondaPacks/

[illegible]

```
du -hc --max-depth 1 ~/MyCondaEnvs/
```

```
531M    /home/aivens2/MyCondaEnvs/snakemake
```

```
531M    /home/aivens2/MyCondaEnvs/
```

531M total

```
du -hc --max-depth 1 ~/MyCondaPacks/
```

```
982M    /home/aivens2/MyCondaPacks/cache
```

```
76M      /home/aivens2/MyCondaPacks/pandas-2.3.3-py313h08cd8bf_1
```

```
40M      /home/aivens2/MyCondaPacks/libopenblas-0.3.30-pthreads h94d23a6 4
```

```
41M      /home/aivens2/MyCondaPacks/numpy-2.3.5-py313hf6604e3_0
```

```
45M /home/aivens2/MyCondaPacks/icu-75.1-he02047a_0
```

```
21M    /home/aivens2/MyCondaPacks/libstdcxx-15.2.0-h934c35e 13
```

```
3.8M /home/aivens2/MyCondaPacks/coin-or-clp-1.17.10-hc03379b_3
```

```
2.5M    /home/aivens2/MyCondaPacks/libsqlite-3.51.0-hee844dc_0
```

```
132M /home/aivens2/MyCondaPacks/python-3.13.9-hc97d973_101_cp313
```

```
6.5M    /home/aivens2/MyCondaPacks/pip-25.3-pyh145f28c_0
```

```
12M      /home/aivens2/MyCondaPacks/libgfortran5-15.2.0-h68bc16d_13
```

• •

```
84K    /home/aivens2/MyCondaPacks/veracitools-0.1.3-py_0
```

```
56K /home/aivens2/MyCondaPacks/typer-slim-standard-0.20.0-h4daf872_1
```

```
36K    /home/aivens2/MyCondaPacks/ libgcc mutex-0.1-conda forge
```

```
1.6G      /home/aivens2/MyCondaPacks/
```

1.6G total

Action time!

(base) aivens2@bioinfmsc8:~\$ conda activate snakemake

(/home/aivens2/MyCondaEnvs/snakemake) aivens2@bioinfmsc8:~\$



Let's run snakemake!

snakemake --version

9.13.7

snakemake

Error: no Snakefile found, tried Snakefile, snakefile, workflow/Snakefile, workflow/snakefile.

Getting help....

snakemake --help

```
usage: snakemake [-h] [--dry-run] [--profile PROFILE] [--workflow-profile WORKFLOW_PROFILE] [--cache [RULE ...]] [--snakefile FILE] [--cores N] [--jobs N]
                  [--local-cores N] [--resources NAME=INT [NAME=INT ...]] [--set-threads RULE=THREADS [RULE=THREADS ...]] [--max-threads MAX_THREADS]
                  [--set-resources RULE=RESOURCE-VALUE [RULE=RESOURCE-VALUE ...]] [--set-scatter NAME=SCATTERITEMS [NAME=SCATTERITEMS ...]]
                  [--set-resource-scopes RESOURCE={global|local} [RESOURCE={global|local} ...]] [--default-resources {NAME=INT ...}]
                  [--preemptible-rules {PREEMPTIBLE_RULES ...}] [--preemptible-retries {PREEMPTIBLE_RETRIES}] [--configfile FILE [FILE ...]] [--config {KEY=VALUE ...}]
                  [--replace-workflow-config] [--envvars VARIABLE [VARIABLE ...]] [--directory DIR] [--touch] [--keep-going]
                  [--rerun-triggers {code,input,mtime,params,software-env} ...] [--code {code,input,mtime,params,software-env} ...] [--force] [--executor {local,dryrun,touch}]
                  [--forceall] [--forusrun {TARGET ...}] [--consider-ancient RULE=INPUTITEMS [RULE=INPUTITEMS ...]] [--prioritize TARGET {TARGET ...}]
                  [--batch RULE=BATCH/BATCHES] [--until TARGET {TARGET ...}] [--omit-from TARGET {TARGET ...}] [--rerun-incomplete] [--shadow-prefix DIR]
                  [--strict-dag-evaluation {cyclic-graph,functions,periodic-wildcards} [{cyclic-graph,functions,periodic-wildcards} ...]] [--scheduler {greedy,slp}]
                  [--conda-base-path {CONDA_BASE_PATH} [--no-subworkflows] [--precommand PRECOMMAND] [--groups GROUPS [GROUPS ...]]
                  [--group-components GROUP COMPONENTS [GROUP COMPONENTS ...]] [--report {FILE}] [--report-after-run] [--report-style {sheet,cssfile}]
                  [--report-metadata FILE] [--reporter {JUGUIN} [--draft-notebook TARGET] [--edit-notebook TARGET] [--notebook-listen IP:PORT] [--lint [{text,json}]]
                  [--generate-unit-tests {TESTPATH}] [--containerize] [--export-cwl FILE] [--list-rules] [--list-target-rules] [--dag [{dot,mermaid-js}]]
                  [--rulegraph [{dot,mermaid-js}] [--filegraph] [--d3dag] [--summary] [--detailed-summary] [--archive FILE] [--cleanup-metadata FILE [FILE ...]]
                  [--cleanup-shadow] [--skip-script-cleanup] [--unlock] [--list-changes {input,code,params}] [--list-input-changes] [--list-params-changes]
                  [--list-untracked] [--delete-all-output] [--delete-temp-output] [--keep-incomplete] [--drop-metadata] [--version] [--printshellcmd] [--debug-dag]
                  [--noquiet] [--quiet [{all,best,progress,reason,rules} ...]] [--print-compile] [--verbose] [--force-use-threads] [--allow-ambiguity] [--noquiet]
                  [--ignore-incomplete] [--max-inventory-time SECONDS] [--trust-io-cache] [--max-checksum-file-size SIZE] [--latency-wait SECONDS]
                  [--wait-for-free-local-storage WAIT_FOR_FREE_LOCAL_STORAGE] [--wait-for-files {FILE ...}] [--wait-for-files-file FILE]
                  [--runtime-source-cache-path PATH] [--queue-input-wait-time SECONDS] [--omit-flags {EXIT_FLAGS [EXIT_FLAGS ...]}] [--notemp] [--all-temp]
                  [--unneeded-temp-files FILE [FILE ...]] [--keep-storage-local-copies] [--not-retrieve-storage] [--target-files-omit-workdir-adjustment]
                  [--allowed-rules ALLOWED_RULES [ALLOWED_RULES ...]] [--max-jobs-per-timespan MAX_JOBS_PER_TIMESPAN]
                  [--max-status-checks-per-second MAX_STATUS_CHECKS_PER_SECOND] [--seconds-between-status-checks SECONDS BETWEEN STATUS CHECKS] [--retries RETRIES]
                  [--wrapper-prefix WRAPPER_PREFIX] [--default-storage-provider DEFAULT_STORAGE_PROVIDER] [--default-storage-prefix DEFAULT_STORAGE_PREFIX]
                  [--local-storage-prefix LOCAL_STORAGE_PREFIX] [--remote-job-local-storage-prefix REMOTE_JOB_LOCAL_STORAGE_PREFIX]
                  [--shared-fs-usage {input-output,persistence,software-deployment,source-cache,sources,storage-local-copies,none}] [{input-output,persistence,software-deployment,source-cache,sources,storage-local-copies,none} ...]]
                  [--scheduler-greediness SCHEDULER_GREEDINESS] [--scheduler-subsample SCHEDULER_SUBSAMPLE] [--no-hooks] [--debug] [--runtime-profile FILE]
                  [--local-groupid LOCAL_GROUPID] [--attempt ATTEMPT] [--show-failed-dir] [--logger {I} [{I} ...]] [--job-deploy-sources] [--benchmark-extended]
                  [--container-image IMAGE] [--jobscript SCRIPT] [--jobscript NAME]
                  [--software-deployment-method {aptinstaller,conda,env-modules} [{aptinstaller,conda,env-modules} ...]] [--container-cleanup-images] [--use-conda]
                  [--conda-not-block-search-path-envvars] [--list-conda-envs] [--conda-prefix DIR] [--conda-cleanup-envs] [--conda-cleanup-pkgs [{tarballs,cache}]]
                  [--conda-unsafe-envvars] [--conda-unsafe] [--use-conda-frontends] [--conda-unsafe] [--use-conda-frontends] [--conda-unsafe] [--conda-unsafe]
                  [--deploy-sources QUERY CHECKSUM] [--target-jobs TARGET_JOBS [TARGET_JOBS ...]] [--mode {default,remote,subprocesses}]
                  [--scheduler-solver-path SCHEDULER_SOLVER_PATH] [--max-jobs-per-second MAX_JOBS_PER_SECOND] [--report-html-path VALUE]
                  [--report-html-style-path VALUE] [--scheduler-greedy-omit-prioritize-by-temp-and-input]
                  [--scheduler-slp-solver VALUE] [--scheduler-slp-solver-path VALUE]
                  [targets ...]

Snakemake is a Python based language and execution environment for GNU Make-like workflows.

options:
  -h, --help            show this help message and exit

EXECUTION:
  targets              Targets to build. May be rules or files.
  --dry-run, --dryrun, -n
                        Do not execute anything, and display what would be done. If you have a very large workflow, use "--dry-run --quiet" to just print a summary of
                        the DAG of jobs. (default: False)

--profile PROFILE      Name of profile to use for configuring Snakemake. Snakemake will search for a corresponding folder in /etc/sd/snakemake and
                        /home/snakemake/alternatively, this can be an absolute or relative path. The profile folder has to contain a file
                        'config.yaml'. This file can be used to set default values for command line options in YAML format. For example, "--cluster qsub" becomes
                        'cluster: qsub' in the YAML file. Profiles can be obtained from https://github.com/snakemake-profiles. The profile can also be set via the
                        environment variable $SNAKEMAKE_PROFILE. To override this variable and use no profile at all, provide the value 'none' to this argument. [env
                        var: SNAKEMAKE_PROFILE]

--workflow-profile WORKFLOW_PROFILE
                        Path (relative to current directory) to workflow specific profile folder to use for configuring Snakemake with parameters specific for this
                        workflow (like resources). If this flag is not used, Snakemake will by default use 'profiles/default' if present (searched both relative to
                        current directory and relative to Snakemake, in this order). For skipping any workflow specific profile provide the special value 'none'.
                        Settings made in the workflow profile will override settings made in the general profile (see "--profile"). The profile folder has to contain a file
                        'config.yaml'. This file can be used to set default values for command line options in YAML format. For example, "--executor slurm" becomes
                        'executor: slurm' in the YAML file. It is advisable to use the workflow profile to set or overwrite e.g. workflow specific resources like the
                        amount of threads for a particular rule or the amount of memory needed. Note that in such cases, the arguments may be given as nested YAML
                        mappings in the profile, e.g. 'set-threads: myrule: 4' instead of 'set-threads: myrule=4'.

--cache [RULE ...]    Store output files of given rules in a central cache given by the environment variable $SNAKEMAKE_OUTPUT_CACHE. Likewise, retrieve output files
                        of the rules from the cache if they have been created before (by anybody writing to the same cache), instead of actually executing the
                        rules. Output files are identified by hashing all steps, parameters and software stack (conda envs or containers) needed to create them. If no
                        rules are given, all rules that are eligible for caching (have a cache directive, see docs) are cached.
                        The workflow definition in form of a snakefile. Usually, you should not need to specify this. By default, Snakemake will search for 'Snakefile',
                        'snakefile', 'workflow/Snakefile', 'workflow/snakefile' beneath the current working directory, in this order. Only if you definitely want a
                        different layout, you need to use this parameter.
                        Use at most N CPU cores/jobs in parallel. If it is omitted or 'all', the limit is set to the number of available CPU cores. In case of
                        cluster/cloud execution, this argument sets the maximum number of cores requested from the cluster or cloud scheduler. (See
                        https://snakemake.readthedocs.io/en/stable/snakefiles/rules.html#resources-remote-execution for more info.) This number is available to rules via
                        workflow.cores.

--cores, -c N          Use at most N CPU cluster/cloud jobs in parallel. For local execution this is an alias for "--cores" (it is though recommended to use "--cores"
                        in that case). Note: Set to 'unlimited' to allow any number of parallel jobs.
                        In cluster/cloud mode, use at most N cores of the host machine in parallel (default: number of CPU cores of the host). The cores are used to
                        execute local rules. This option is ignored when not in cluster/cloud mode.

--local-cores N         Define additional resources that shall constrain the scheduling analogously to "--cores" (see above). A resource is defined as a name and an
                        integer value. E.g. "--resources mem_mb=1000". Rules can use resources by defining the resource keyword, e.g. 'resources: mem_mb=600'. If now two
                        rules require 600 of the resource 'mem_mb' they won't be run in parallel by the scheduler. In cluster/cloud mode, this argument will also
                        constrain the amount of resources requested from the server. (See https://snakemake.readthedocs.io/en/stable/snakefiles/rules.html#resources-
                        remote-execution for more info.) (default: )

--set-threads RULE=THREADS [RULE=THREADS ...]
                        Overwrite thread usage of rules. This allows to fine-tune workflow parallelisation. In particular, this is helpful to target certain cluster
                        nodes by e.g. shifting a rule to use more, or less threads than defined in the workflow. Thereby, THREADS has to be a positive integer, and RULE
                        has to be the name of the rule. (default: )

--max-threads MAX_THREADS
                        Define a global maximum number of threads available to any rule. Rules requesting more threads (via the threads keyword) will have their values
                        reduced to the maximum. This can be useful when you want to restrict the maximum number of threads without modifying the workflow definition or
                        overwriting rules individually with "--set-threads".

--set-resources RULE=RESOURCE-VALUE [RULE=RESOURCE-VALUE ...]
                        Overwrite resource usage of rules. This allows to fine-tune workflow resources. In particular, this is helpful to target certain cluster nodes by
                        e.g. defining a certain partition for a rule or overwriting a temporary directory. Thereby, RESOURCE-VALUE has to be a positive integer or a string, RULE
                        has to be the name of the rule, and RESOURCE has to be the name of the resource. (default: )

--set-scatter NAME=SCATTERITEMS [NAME=SCATTERITEMS ...]
                        Overwrite number of scatter items of scattergather processes. This allows to fine-tune workflow parallelisation. Thereby, SCATTERITEMS has to be
                        a positive integer, and NAME has to be the name of the scattergather process defined via a scattergather directive in the workflow. (default: )

--set-resource-scopes RESOURCE={global|local} [RESOURCE={global|local} ...]
                        Overwrite resource scopes. A scope determines how a constraint is reckoned in cluster execution. With RESOURCE=local, a constraint applied to
                        RESOURCE using "--resources" will be considered the limit for each group submission. With RESOURCE=global, the constraint will apply across all
                        groups cumulatively. By default, only 'mem_mb' and 'disk_mb' are considered local, all other resources are global. This may be modified in the
                        snakefile using the 'resource_scopes' directive. Note that number of threads, specified via "--cores", is always considered local. (See
                        https://snakemake.readthedocs.io/en/stable/snakefiles/rules.html#resources-remote-execution for more info) (default: )

--default-resources, --default-res {NAME=INT ...}
                        Define default values of resources for rules that do not define their own values. In addition to plain integers, python expressions over
                        inputsize are allowed (e.g. '2*input.size.mb'). The inputsize is the sum of the sizes of all input files of a rule. By default, Snakemake assumes
                        a default for mem_mb, disk_mb, and tmpdir (see below). This option allows to add further defaults (e.g. account and partition for slurm) or to
                        overwrite these default values. The defaults are 'mem_mb=main(max(2*input.size.mb, 1000), 8000)', 'disk_mb=main(2*input.size.mb, 1000)' (i.e.,
                        default disk and mem usage is twice the input file size but at least 1GB), and the system temporary directory (as given by TMPDIR, TEMP, or
                        TMP) is used for the tmpdir resource. The tmpdir resource is automatically used by shell commands, scripts and wrappers to store temporary data
                        as it is mirrored into TMPDIR, TEMP, and TMP for the executed subprocesses). If this argument is not specified at all, Snakemake just uses
                        the tmpdir resource as outlined above. The tmpdir resource can also be overwritten in the same way as e.g. mem_mb above. Thereby, it is even
                        possible to use shutil.disk_usage(system.tmpdir) free and comparing this to input.size in order to determine if one can expect the system tmpdir
                        to be big enough and switch to another tmpdir in case it is not.

--preemptible-rules {PREEMPTIBLE_RULES ...}
                        Define which rules shall use a preemptible machine which can be prematurely killed by e.g. a cloud provider (also called spot instances). This is
                        currently only supported by the Google Life Sciences executor and ignored by all other executors. If no rule names are provided, all rules are
                        considered to be preemptible.

--preemptible-retries {PREEMPTIBLE_RETRIES}
                        Number of retries that shall be made in order to finish a job from rules that has been marked as preemptible via the --preemptible-rules
                        setting.

--configfile, --configfiles FILE [FILE ...]
                        Specify or overwrite the config file of the workflow (see the docs). Values specified in JSON or YAML format are available in the global config
                        dictionary inside the workflow. Multiple files overwrite each other in the given order. Thereby missing keys in previous config files are
                        extended by following configfiles. Note that this order also includes a config file defined in the workflow definition itself (which will come
                        first). (default: )

--config, -C {KEY=VALUE ...}
                        Set or overwrite values in the workflow config object. The workflow config object is accessible as variable config inside the workflow. Default
                        values can be set by providing a YAML JSON file (see "--configfile" and documentation).

--replace-workflow-config
                        Config files provided via command line do not update and extend the config dictionary of the workflow but instead fully replace it. Keys that are
                        not defined in the provided config files will be undefined even if specified within the workflow config. (default: False)

--envvars VARIABLE [VARIABLE ...]
                        Environment variables to pass to cloud jobs.

--directory, -d DIR    Specify working directory (relative paths in in the snakefile will use this as their origin).

--touch, -t            Touch output files (mark them up to date without really changing them) instead of running their commands. This is used to pretend that the rules
                        were executed, in order to fool future invocations of snakemake. Fails if a file does not yet exist. Note that this will only touch files that
                        would otherwise be recreated by Snakemake (e.g., because their input files are newer). For enforcing a touch, combine this with "--force",
                        "--forceall", or "--forusrun". Note however that you lose the provenance information when the files have been created in reality. Hence, this
                        should be used only as a last resort. (default: False)

--rerun-triggers {code,input,mtime,params,software-env} ...
                        Do so with independence, i.e. if a job fails during execution. This only applies to runtime failures in job execution, not to errors during
                        workflow parsing or DAG construction. (default: False)

--force, -f            Force the execution of the selected target or the first rule regardless of already created output. (default: False)

--executor, -e {local,dryrun,touch}
                        Specify a custom executor, available via an executor plugin: snakemake_executor_

--forceall, -F         Force the execution of the selected (or the first) rule and all rules it is dependent on regardless of already created output. (default: False)

--forusrun, -R [TARGET ...]
                        Force the execution of the selected (or the first) rule and all rules it is dependent on regardless of already created output. (default: False)
```

```
Force the re-execution or creation of the given rules or files. Use this option if you changed a rule and want to have all its output in your workflow updated.
--consider-ancient RULE-INPUTITEMS [RULE-INPUTITEMS ...]
    Consider given input items of given rules as ancient, i.e. not triggering re-runs if they are newer than the output files. Putting this into a workflow specific profile (or specifying as argument) allows to overrule rerun triggers caused by file modification dates where the user knows better. RULE is the name of the rule, INPUTITEMS is a comma separated list of input items of the rule (given as name or index (0-based)).
    default: {}
--prioritize, -P TARGET [TARGET ...]
    Tell the scheduler to assign creation of given targets (and all their dependencies) highest priority.
--batch RULE-BATCH/RATCHES
    Only create the given BATCH of the input files of the given RULE. This can be used to iteratively run parts of very large workflows. Only the execution plan of the relevant part of the workflow has to be calculated, thereby speeding up DAG computation. It is recommended to provide the most suitable rule for batching when documenting a workflow. It should be some aggregating rule that would be executed only once, and has a large number of input files. For example, it can be a rule that aggregates over samples.
--until, -U TARGET [TARGET ...]
    Run the pipeline until it reaches the specified rules or files. Only runs jobs that are dependencies of the specified rule or files, does not run sibling DAGs.
--omit-from, -O TARGET [TARGET ...]
    Prevent the execution or creation of the given rules or files as well as any rules or files that are downstream of these targets in the DAG. Also runs jobs in sibling DAGs that are independent of the rules or files specified here.
--rerun-incomplete, --ri
    Re-run all jobs the output of which is recognized as incomplete. (default: False)
--shadow-prefix DIR
    Specify a directory in which the 'shadow' directory is created. If not supplied, the value is set to the '.snakemake' directory relative to the working directory.
--strict-dag-evaluation [(cyclic-graph,functions,periodic-wildcards)] [(cyclic-graph,functions,periodic-wildcards) ...]
    Strict evaluation of rules' correctness even when not required to produce the output files.
--scheduler [(greedy,ilp)]
    Specifies the scheduling plugin to use. Builtin plugins are greedy (fast) and ilp, while the latter scheduler aims to reduce runtime and hold usage by best possible use of resources. (default: ilp)
--conda-base-path CONDA_BASE_PATH
    Path of conda base installation (home of conda, mamba, activate) (internal use only).
--no-subworkflows, --nsw
    Do not evaluate or execute subworkflows. (default: False)
--precommand PRECOMMAND
    Only used in case of remote execution. Command to be executed before Snakemake executes each job on the remote compute node.
--dag [(dot,mermaid-js)]
    Do not execute anything and print the directed acyclic graph of jobs in the dot language or in mermaid-js. Recommended use on this systems: 'snakemake --dag | dot | display'. Note print statements in your Snakefile may interfere with visualization.
GROUPING:
--groups GROUPS [GROUPS ...]
    Assign rules to groups (this overwrites any group definitions from the workflow). (default: {})
--group-components GROUP_COMPONENTS [GROUP_COMPONENTS ...]
    Set the number of connected components a group is allowed to span. By default, this is 1, but this flag allows to extend this. This can be used to run e.g. 3 jobs of the same rule in the same group, although they are not connected. It can be helpful for putting together many small jobs or benefitting of shared memory setups. (default: 1)
REPORTS:
--report [FILE]
    Create a self-contained HTML report with default statistics, provenance information and user-specified results. For smaller datasets with a limited report complexity, you can specify an '.html' file and all results will be embedded directly into this file. For customized reports on larger sample sizes, it makes more sense to specify a '.sip' file. The resulting archive will spread the contents across a folder structure, for a quicker loading of individual results. You can unpack this archive anywhere and open the 'report.html' file in its main folder to view the report in any web browser.
--report-after-run
    After finishing the workflow, directly create the report. It is required to provide --report. (default: False)
--report-style-sheet CSSFILE
    Custom stylesheet to use for report. In particular, this can be used for branding the report with e.g. a custom logo, see docs.
--report-metadata FILE
    Custom metadata to use for the landing page of the report. In particular, this can be used to provide metadata in the report e.g. the work directory, see docs.
--reporter PLUGIN
    Specify a custom report plugin. By default, Snakemake's builtin html reporter will be used. For custom reporters, check out their command line options starting with '--report-'.
NOTEBOOKS:
--draft-notebook TARGET
    Draft a skeleton notebook for the rule used to generate the given target file. This notebook can then be opened in a jupyter server, executed and implemented until ready. After saving, it will automatically be reused in non-interactive mode by Snakemake for subsequent jobs.
--edit-notebook TARGET
    Interactively edit the notebook associated with the rule used to generate the given target file. This will start a local jupyter notebook server. Any changes to the notebook should be saved, and the server has to be stopped by closing the notebook and hitting the 'Quit' button on the jupyter dashboard. Afterwards, the updated notebook will be automatically stored in the path defined in the rule. If the notebook is not yet present, this will create an empty draft.
--notebook-listen IP:PORT
    The IP address and PORT the notebook server used for editing the notebook ('--edit-notebook') will listen on. (default: localhost:8888)
UTILITIES:
--lint [{text,json}]
    Perform linting on the given workflow. This will print snakemake specific suggestions to improve code quality (work in progress, more lints to be added in the future). If no argument is provided, plain text output is used.
--generate-unit-tests [TESTPATH]
    Automatically generate unit tests for each workflow rule. This assumes that all input files of each rule are already present. Jobs without present input files will be skipped (a warning will be issued). For each rule, one test case will be created and, after successful execution, tests can be run with pytest TESTPATH.
--containerize
    Print a Dockerfile that provides an execution environment for the workflow, including all conda environments. (default: False)
--export-od FILE
    Compile workflow to OM and store it in given FILE.
--list-rules, --list, -l
    Show available rules in given Snakefile. (default: False)
--list-target-rules, --lt
    Show available target rules in given Snakefile. (default: False)
--rulegraph [(dot,mermaid-js)]
    Do not execute anything and print the dependency graph of rules in the dot language or in mermaid-js. This will be less crowded than above DAG of jobs, but also show less information. Note that each rule is displayed once, hence the displayed graph will be cyclic if a rule appears in several steps of the workflow. Use this if above option leads to a DAG that is too large. Recommended use on Unix systems: snakemake --rulegraph | dot | display. Note print statements in your Snakefile may interfere with visualization.
--filegraph
    Do not execute anything and print the dependency graph of rules with their input and output files in the dot language. This is an intermediate solution between above DAG of jobs and the rule graph. Note that each rule is displayed once, hence the displayed graph will be cyclic if a rule appears in several steps of the workflow. Use this if above option leads to a DAG that is too large. Recommended use on Unix systems: snakemake --filegraph | dot | display. Note print statements in your Snakefile may interfere with visualization. (default: False)
--d3dag
    Print the DAG in D3.js compatible JSON format. (default: False)
--summary, -s
    Print a summary of all files created by the workflow. The has the following columns: filename, modification time, rule version, status, plan. Thereby rule version contains the version the file was created with (see the version keyword of rules), and status denotes whether the file is missing, its input files are newer or if version or implementation of the rule changed since file creation. Finally the last column denotes whether the file will be updated or created during the next workflow execution. (default: False)
--detailed-summary, -D
    Print a summary of all files created by the workflow. The has the following columns: filename, modification time, rule version, input file(s), shell command, status, plan. Thereby rule version contains the version the file was created with (see the version keyword of rules), and status denotes whether the file is missing, its input files are newer or if version or implementation of the rule changed since file creation. The input file and shell command columns are self explanatory. Finally the last column denotes whether the file will be updated or created during the next workflow execution. (default: False)
--archive FILE
    Archive the workflow into the given tar archive FILE. The archive will be created such that the workflow can be re-executed on a vanilla system. The function needs conda and git to be installed. It will archive every file that is under git version control. Note that it is best practice to save the Snakefile, config files, and scripts under version control. Hence, they will be included in the archive. Further, it will add input files that are not generated by the workflow itself and conda environments. Note that symlinks are dereferenced. Supported formats are '.tar', '.tar.gz', '.tar.bz2' and '.tar.xz'.
--cleanup-metadata, --cm FILE [FILE ...]
    Cleanup the metadata of given files. That means that snakemake removes any tracked version info, and any marks that files are incomplete.
--cleanup-shadow
    Cleanup old shadow directories which have not been deleted due to failures or power loss. (default: False)
--skip-script-cleanup
    Don't delete wrapper scripts used for execution (default: False)
--unlock
    Remove a lock on the working directory. (default: False)
--list-changes, --lc [(input,code,params)]
    List all output files for which the given items (code, input, params) have changed since creation.
--list-input-changes, --li
    List all output files for which the defined input files have changed in the Snakefile (e.g. new input files were added in the rule definition or files were renamed). For listing input file modification in the filesystem, use '--summary'. (default: False)
--list-params-changes, --lp
    List all output files for which the defined params have changed in the Snakefile. (default: False)
--list-untracked, --lu
    List all files in the working directory that are not used in the workflow. This can be used e.g. for identifying leftover files. Hidden files and directories are ignored. (default: False)
--delete-all-output
    Remove all files generated by the workflow. Use together with '--dry-run' to list files without actually deleting anything. Note that this will not recurse into subworkflows. Write-protected files are not removed. Nevertheless, use with care! (default: False)
--delete-temp-output
    Remove all temporary files generated by the workflow. Use together with '--dry-run' to list files without actually deleting anything. Note that this will not recurse into subworkflows. (default: False)
--keep-incomplete
    Do not remove incomplete output files by failed jobs. (default: False)
--drop-metadata
    Drop metadata file tracking information after job finishes. Provenance-information based reports (e.g. '--report' and the '--list_x_changes' functions) will be empty or incomplete. (default: False)
--version, -v
    show program's version number and exit
OUTPUT:
--printshellcmds, -p
    Print out the shell commands that will be executed. (default: False)
--debug-dag
    Print candidate and selected jobs (including their wildcards) while inferring DAG. This can help to debug unexpected DAG topology or errors. (default: False)
--nocolor
    Do not use a colored output. (default: False)
--quiet, -q [(all,host,progress,reason,rules)] ...]
    Do not output certain information. If used without arguments, do not output any progress or rule information. Defining 'all' results in no information being printed at all.
--print-compile
    Print the python representation of the workflow. (default: False)
--verbose
    Print debugging output. (default: False)
BEHAVIOR:
--force-use-threads
    Force threads rather than processes. Helpful if shared memory (/dev/shm) is full or unavailable. (default: False)
--allow-ambiguity, -a
    Don't check for ambiguous rules and simply use the first if several can produce the same file. This allows the user to prioritize rules by their order in the snakefile. (default: False)
--nolock
    Do not lock the working directory (default: False)
--ignore-incomplete, --ii
    Do not check for incomplete output files. (default: False)
--max-inventory-time SECONDS
    Spend at most SECONDS seconds to create a file inventory for the working directory. The inventory vastly speeds up file modification and existence checks when computing which jobs need to be executed. However, creating the inventory itself can be slow, e.g. on network file systems. Hence, we do not spend more than a given amount of time and fall back to individual checks for the rest. (default: 20)
--trust-io-cache
    Tell Snakemake to assume that all input and output file existence and modification time queries performed in previous dryruns are still valid and therefore don't have to be repeated. This can lead to speed-ups, but implies that input and output have not been modified manually in between. Non dry-run execution will automatically invalidate the cache and lead to redoing the queries. (default: False)
--max-checksum-file-size SIZE
    Compute the checksum during DAG computation and job postprocessing only for files that are smaller than the provided threshold (given in any valid size unit, e.g. MB, which is also the default). (default: 1000000)
--latency-wait, -w SECONDS
    Wait given seconds if an output file of a job is not present after the job finished. This helps if your filesystem suffers from latency. (default: 5)
--wait-for-free-local-storage WAIT_FOR_FREE_LOCAL_STORAGE
    Wait for given timespan for enough free local storage when downloading remote storage files. If not set, no waiting is performed.
--wait-for-files [FILE ...]
    Wait '--latency-wait' seconds for these files to be present before executing the workflow. This option is used internally to handle filesystem latency in cluster environments.
--wait-for-files-file FILE
    Same behaviour as '--wait-for-files', but file list is stored in file instead of being passed on the commandline. This is useful when the list of files is too long to be passed on the commandline. Meant for internal use.
--runtime-source-cache-path PATH
    Path to the runtime source cache directory. Meant for internal use.
--queue-input-wait-time SECONDS
    Set the interval in seconds to check for new input in rules that use from_queue to obtain input files. (default: 10)
--omit-flags OMIT_FLAGS [OMIT_FLAGS ...]
    Omit the given input and output file flags (e.g. pipe). This can be useful for debugging.
```

```
--notemp, --notemp, --nt
    Ignore temp() declarations. This is useful when running only a part of the workflow, since temp() would lead to deletion of probably needed files
    by other parts of the workflow. (default: False)
--all-temp
    Mark all output files as temp files. This can be useful for CI testing, in order to save space. (default: False)
--unneeded-temp-files FILE [FILE ...]
    Given files will not be uploaded to storage and immediately deleted after job or group job completion.
--keep-storage-local-copies
    Keep local copies of remote input and output files. (default: False)
--not-retrieve-storage
    Do not retrieve remote files (default is to retrieve remote files). (default: False)
--target-files-omit-workdir-adjustment
    Do not adjust the paths of given target files relative to the working directory. (default: False)
--allowed-rules ALLOWED_RULES [ALLOWED_RULES ...]
    Only consider given rules. If omitted, all rules in Snakefile are used. Note that this is intended primarily for internal use and may lead to
    unexpected results otherwise. Meant for internal use or debugging.
--max-jobs-per-timespan MAX_JOBS_PER_TIMESPAN
    Maximal number of job submissions/executions per timespan. Format: , e.g. 50/1m or 0.5/1s. (default: 100/1s)
--max-status-checks-per-second MAX_STATUS_CHECKS_PER_SECOND
    Maximal number of job status checks per second/ fractions allowed. (default: 10)
--seconds-between-status-checks SECONDS_BETWEEN_STATUS_CHECKS
    Number of seconds to wait between two rounds of status checks. (default: 10)
--retries, --restart-times, -F RETRIES
    Number of times to restart failing jobs. (default: 0)
--wrapper-prefix WRAPPER_PREFIX
    URL prefix for wrapper directive. Set this to use your fork or a local clone of the repository, e.g., use a git URL like
    gitfile://path/to/your/local/clone/. (default: https://github.com/snakemake/snakemake-wrappers/raw/)
--default-storage-provider DEFAULT_STORAGE_PROVIDER
    Specify default storage provider to be used for all input and output files that don't yet specify one (e.g., "s3"). See
    https://snakemake.github.io/snakemake-plugin-catalog for available storage provider plugins. If not set or explicitly 'none', no default storage
    provider will be used.
--default-storage-prefix DEFAULT_STORAGE_PREFIX
    Specify prefix for default storage provider. E.g. a bucket name.
--local-storage-prefix LOCAL_STORAGE_PREFIX
    Specify prefix for storing local copies of storage files and folders (e.g. local scratch disk). Environment variables will be expanded. (default:
    snakemake/storage)
--remote-job-local-storage-prefix REMOTE_JOB_LOCAL_STORAGE_PREFIX
    Specify prefix for storing local copies of storage files and folders (e.g. local scratch disk) in case of remote jobs (e.g. cluster or cloud
    jobs). Environment variables will be expanded within the remote job. (default: snakemake/storage)
--shared-fs-usage (input-output,persistence,software-deployment,source-cache,sources,storage-local-copies,none) [(input-output,persistence,software-deployment,source-cache,sources,storage-local-copies,none) ...]
    Set assumptions on shared filesystem for non-local workflow execution. To disable any sharing via the filesystem, specify 'none'. Usually, the
    executor plugin sets this to a correct default. However, sometimes it is worth tuning this setting, e.g. for optimizing cluster performance. For
    example, when using "--default-storage-provider fs" together with a cluster executor like slurm, you might want to set "--shared-fs-usage
    persistence software-deployment sources source-cache", such that software deployment and data provenance will be handled by NFS but input and
    output files will be handled exclusively by the storage provider. (default: input-output persistence software-deployment source-cache sources
    storage-local-copies)
--scheduler-greediness, --greediness SCHEDULER_GREEDINESS
    Set the greediness of scheduling. This value between 0 and 1 determines how careful jobs are selected for execution. The default value (1.0)
    provides the best speed and still acceptable scheduling quality. Deprecated in favor of "--scheduler-greedy-greediness". (default: 1.0)
--scheduler-subsample SCHEDULER_SUBSAMPLE
    Set the number of jobs to be considered for scheduling. If number of ready jobs is lower, this option has no effect. This can be useful on very large DAGs, where the scheduler
    can take some time selecting which jobs to run.
--no-hooks
    Do not invoke onstart, onsuccess or onerror hooks after execution. (default: False)
--debug
    Allow to debug rules with e.g. PDB. This flag allows to set breakpoints in run blocks. (default: False)
--runtime-profile FILE
    Profile Snakemake and write the output to FILE. This requires yappi to be installed.
--local-groupid LOCAL_GROUPID
    Internal use only: Name for local groupid. (default: local)
--attempt ATTEMPT
    Internal use only: define the initial value of the attempt parameter. (default: 1)
--show-failed-logs
    Automatically display logs of failed jobs. (default: False)
--logger {} [{} ...]
    Specify one or more custom loggers, available via logger plugins. (default: )
--job-deploy-sources
    Whether the workflow sources shall be deployed before a remote job is started. Only applies if "--no-shared-fs" is set or executors are used that
    imply no shared FS (e.g. the Kubernetes executor). (default: False)
--benchmark-extended
    Write extended benchmarking metrics. (default: False)

REMOTE EXECUTION:
--container-image IMAGE
    Docker image to use, e.g., when submitting jobs to Kubernetes. Defaults to https://hub.docker.com/r/snakemake/snakemake, tagged with the same
    version as the currently running Snakemake instance. Note that overwriting this value is up to your responsibility. Any used image has to contain
    a working snakemake installation that is compatible with (or ideally the same as) the currently running version. (default:
    snakemake/snakemake:v1.13.7)
--immediate-submit, --is
    Immediately submit all jobs to the cluster instead of waiting for present input files. This will fail, unless you make the cluster aware of job
    dependencies, e.g. via: $ snakemake --cluster 'shatch --dependency (dependencies)'. Assuming that your submit script (here shatch) outputs the
    generated job id to the first stdout line, (dependencies) will be filled with space separated job ids this job depends on. Does not work for
    workflows that contain checkpoint rules, and localrules will be skipped. The additional argument "--notemp" should be specified. Most often,
    "--not-retrieve-storage" is also recommended to avoid Snakemake trying to download output files before the jobs producing them are executed.
    (default: False)
--jobscript, --js SCRIPT
    Provide a custom job script for submission to the cluster. The default script resides as 'jobscript.sh' in the installation directory.
--jobname, --jn NAME
    Provide a custom name for the jobscript that is submitted to the cluster (see "--cluster"). The wildcard '{jobid}' has to be present in the name.
    (default: snakejob.{name}.{jobid}.sh)

SOFTWARE DEPLOYMENT:
--software-deployment-method, --deployment-method, --deployment, --adm (aptainer,conda,env-modules) [(aptainer,conda,env-modules) ...]
    Specify software environment deployment method.
--container-cleanup-images
    Remove unused containers (default: False)

CONDA:
--use-conda
    If defined in the rule, run job in a conda environment. If this flag is not set, the conda directive is ignored. (default: False)
--conda-not-block-search-path-envvars
    Do not block environment variables that modify the search path (R_LIBS, PYTHONPATH, PERLLIB, PERLLIB) when using conda environments. (default:
    False)
--list-conda-envs
    List all conda environments and their location on disk. (default: False)
--conda-prefix DIR
    Specify a directory in which the 'conda' and 'conda-archive' directories are created. These are used to store conda environments and their
    archives, respectively. If not supplied, the value is set to the '.snakemake' directory relative to the invocation directory. If supplied, the
    "--use-conda" flag must also be set. The value may be given as a relative path, which will be extrapolated to the invocation directory, or as an
    absolute path. The value can also be provided via the environment variable $SNAKEMAKE_CONDA_PREFIX. In any case, the prefix may contain
    environment variables which will be properly expanded. Note that if you use remote execution e.g. on a cluster and you have node specific values
    for this, you should disable assuming shared fs for software-deployment (see "--shared-fs-usage").
--conda-cleanup-envs
    Cleanup unused conda environments. (default: False)
--conda-cleanup-pkgs [(tarballs,cache)]
    Cleanup conda packages after creating environments. In case of 'tarballs' mode, will clean up all downloaded package tarballs. In case of 'cache'
    mode, will additionally clean up unused package caches. (default: tarballs)
--conda-creates-envs-only
    If specified, only creates job-specific conda environments then exits. The "--use-conda" flag must also be set. (default: False)
--conda-frontend (conda,mamba)
    Choose the conda frontend for installing environments. (default: conda)

APTAINER/SINGULARITY:
--use-aptainer, --use-singularity
    If defined in the rule, run job within a aptainer/singularity container. If this flag is not set, the singularity directive is ignored.
    (default: False)
--aptainer-prefix, --singularity-prefix DIR
    Specify a directory in which aptainer/singularity images will be stored. If not supplied, the value is set to the '.snakemake' directory relative
    to the invocation directory. If supplied, the "--use-aptainer" flag must also be set. The value may be given as a relative path, which will be
    extrapolated to the invocation directory, or as an absolute path. If not supplied, APTAINER_CACHEDIR is used. In any case, the prefix may
    contain environment variables which will be properly expanded. Note that if you use remote execution e.g. on a cluster and you have node specific
    values for this, you should disable assuming shared fs for software-deployment (see "--shared-fs-usage").
--aptainer-args, --singularity-args ARGS
    Pass additional args to aptainer/singularity.

ENVIRONMENT MODULES:
--use-envmodules
    If defined in the rule, run job within the given environment modules, loaded in the given order. This can be combined with "--use-conda" and
    "--use-singularity", which will then be only used as a fallback for rules which don't define environment modules. (default: False)

INTERNAL:
--deploy-sources QUERY CHECKSUM
    Internal use only: Deploy sources archive from given storage provider query to the current working subdirectory and control for archive checksum
    to proceed.
--target-jobs TARGET_JOBS [TARGET_JOBS ...]
    Internal use only: Target particular jobs by RULE,WILDCARD1=VALUE,WILDCARD2=VALUE,...
--mode (default,remote,subprocess)
    Internal use only: Set execution mode of Snakemake. (default: default)
--scheduler-solver-path SCHEDULER_SOLVER_PATH
    Internal use only: Set the PATH to search for scheduler solver binaries. Deprecated, use --scheduler-rlp-solver-path instead.

DEPRECATED:
--max-jobs-per-second MAX_JOBS_PER_SECOND
    Maximal number of job submissions/executions per second. Deprecated in favor of "--max-jobs-per-timespan".

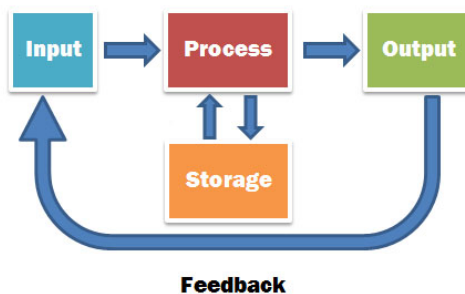
html report plugin settings:
--report-html-path VALUE
    Path to the report file (either .html or .zip). Use zip if your report contains large results or directories with htmlindex as results.
--report-html-stylesheet-path VALUE
    Path to a custom stylesheet for the report.

greedy scheduler plugin settings:
--scheduler-greedy-greediness VALUE
    Set the greediness of scheduling. This value between 0 and 1 determines how careful jobs are selected for execution. The default value (1.0)
    provides the best speed and still acceptable scheduling quality.
--scheduler-greedy-omit-prioritize-by-temp-and-input
    If set, jobs with larger temporary or input files are not prioritised. The rationale of the prioritization is that temp files should be removed
    as soon as possible, and larger input files may take longer to process, so it is better to start them earlier.

ilp scheduler plugin settings:
--scheduler-rlp-solver VALUE
    Set MILP solver to use
--scheduler-rlp-solver-path VALUE
    Set the PATH to search for scheduler solver binaries.

In general, command-line values override environment variables which override defaults.
```

📌 How does it work in theory?



snakemake is **ALL ABOUT MAKING "TARGETS"**

1. snakemake reads a snakefile
2. a snakefile defines a set of rules
3. rules have inputs and outputs and actions
4. snakemake works out the correct order of rules to reach a given target, then runs what is in the rules.

Stage 1

- Read through snakefile
- Evaluate all bare statements
- Collect rules
- Starting from target(s): the "final" output

Stage 2

- Work backwards by matching rule outputs
- Check input files exist
- Generate "DAG"

Stage 3

- Execute the "DAG"
- Run jobs for which all inputs are available
- Continue until target is reached

🏠Common reasons for failure...

Stage 1

- Syntax errors such as missing commas, unbalanced quotes or braces
- Wrong indentation
- Failure to evaluate expressions outside of rules as part of input or output
- Other problems with rule definitions e.g. conflicting wildcards

Stage 2

- Failure to determine targets
- Multiple rules can make the same output (you can set priorities if this was intentional)
- No rule to make a target file (generally the hardest to debug :- ()
- Circular dependency found
- An output file is write-protected : you just don't have permission to create the file!

Stage 3

- If a job fails, snakemake will report an error, and **delete all output files** for that job
- A job fails if anything inside the `{` curly brackets `}` is not valid
- Any shell command returns non-zero status
- You reference an unset `$shell_variable`
- Any output file is not created
- Concurrent jobs will be allowed to finish



The files we are going to use are here:

```
/localdisk/data/BPSM/Lecture22
```

Our first example set uses four files:

```
ds1.csv ds2.csv myplotter.py Snakefile1
```

This time, you should copy them as required to a suitable place in your git repository area (suggested location `${HOME}/Exercises/Lecture22`) as you may want to modify their contents...

The first two are comma separated values (CSV) files containing some "biological" data:

```
head *.csv
```

```
==> ds1.csv <==  
Animal,X,Y  
interesting_bees,8,7  
boring_bees,10,12  
interesting_turtles,2,1  
boring_turtles,0,0  
interesting_ducks,10,9  
boring_ducks,13,7  
interesting_swans,18,16  
boring_swans,16,5  
  
==> ds2.csv <==  
Animal,X,Y  
interesting_salmon,18,2  
boring_salmon,12,10  
interesting_grebes,18,1  
boring_grebes,0,0  
interesting_weasels,3,9  
boring_weasels,13,73  
interesting_rabbits,18,7  
boring_rabbits,16,8  
interesting_salamanders,12,6
```

The "myplotter.py" script is able to plot these type of files and output PDF documents.

```
# Run the script
```

```
./myplotter.py -o plots.pdf ds1.csv
```

```
Traceback (most recent call last):
```

```
  File "/localdisk/home/aivens2/Exercises/Lecture22/./myplotter.py", line 5, in <module>  
    import matplotlib
```

```
ModuleNotFoundError: No module named 'matplotlib'
```

ISSUE: it says there is no `matplotlib` library in this environment!

```
# The script...
```

```
cat myplotter.py
```

```
#!/usr/bin/env python3  
import pandas as pd  
import matplotlib  
matplotlib.use('agg')  
import matplotlib.pyplot as plt
```



```
from matplotlib.backends.backend_pdf import PdfPages

from argparse import ArgumentParser
a = ArgumentParser()
a.add_argument("-o", "--output")
a.add_argument("input", nargs='+')
args = a.parse_args()

dfs = [ pd.read_csv(f) for f in args.input ]
with PdfPages(args.output, keep_empty=False) as pdf:
    for df in dfs:
        df.plot(kind="scatter", x='X', xlim=(0,df['X'].max()+2),
                    y='Y', ylim=(0,df['Y'].max()+2))

        for _, row in df.iterrows():
            plt.annotate(row['Animal'], (row['X'],row['Y']))
        plt.subplots_adjust(left=0.2, right=0.8)
        pdf.savefig()
        plt.close()
```

We need matplotlib and pandas

pip install matplotlib pandas

```
Collecting matplotlib
  Downloading matplotlib-3.10.7-cp313-cp313-manylinux2014_x86_64.manylinux_2_17_x86_64.whl
Requirement already satisfied: pandas in /home/aivens2/MyCondaEnvs/snakemake/lib/python3.
Collecting contourpy>=1.0.1 (from matplotlib)
  Downloading contourpy-1.3.3-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl
Collecting cycler>=0.10 (from matplotlib)
  Downloading cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.22.0 (from matplotlib)
  Downloading fonttools-4.60.1-cp313-cp313-manylinux1_x86_64.manylinux2014_x86_64.manylinu
Collecting kiwisolver>=1.3.1 (from matplotlib)
  Downloading kiwisolver-1.4.9-cp313-cp313-manylinux2014_x86_64.manylinux_2_17_x86_64.whl
Requirement already satisfied: numpy>=1.23 in /home/aivens2/MyCondaEnvs/snakemake/lib/pyth
Requirement already satisfied: packaging>=20.0 in /home/aivens2/MyCondaEnvs/snakemake/lib/
Collecting pillow>=8 (from matplotlib)
  Downloading pillow-12.0.0-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.me
Requirement already satisfied: pyparsing>=3 in /home/aivens2/MyCondaEnvs/snakemake/lib/pyt
Requirement already satisfied: python-dateutil>=2.7 in /home/aivens2/MyCondaEnvs/snakemake
Requirement already satisfied: pytz>=2020.1 in /home/aivens2/MyCondaEnvs/snakemake/lib/pyt
Requirement already satisfied: tzdata>=2022.7 in /home/aivens2/MyCondaEnvs/snakemake/lib/p
Requirement already satisfied: six>=1.5 in /home/aivens2/MyCondaEnvs/snakemake/lib/python3
Download matplotlib-3.10.7-cp313-cp313-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (
      8.7/8.7 MB 18.5 MB/s
Download contourpy-1.3.3-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (3
Download cycler-0.12.1-py3-none-any.whl (8.3 kB)
Download fonttools-4.60.1-cp313-cp313-manylinux1_x86_64.manylinux2014_x86_64.manylinux
      4.9/4.9 MB 39.9 MB/s
Download kiwisolver-1.4.9-cp313-cp313-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (1
      1.5/1.5 MB 27.7 MB/s
Download pillow-12.0.0-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (7.0
      7.0/7.0 MB 47.7 MB/s
Installing collected packages: pillow, kiwisolver, fonttools, cycler, contourpy, matplotli
Successfully installed contourpy-1.3.3 cycler-0.12.1 fonttools-4.60.1 kiwisolver-1.4.9 mat
```

Try to run the script

./myplotter.py -o plots.pdf ds1.csv

```
/localdisk/home/aivens2/Exercises/Lecture22/./myplotter.py:19:
MatplotlibDeprecationWarning: The 'keep_empty' parameter of __init__() was deprecated
in Matplotlib 3.10 and will be removed in 3.12. This parameter does nothing. If any
parameter follows 'keep_empty', they should be passed as keyword, not positionally.
with PdfPages(args.output, keep_empty=False) as pdf:
```

View the plots.pdf file

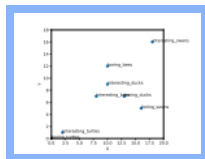
evince plots.pdf &

plots.pdf

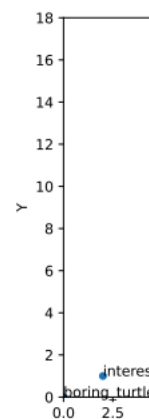
1 / 1

80%

+



1



```
snakemake --cores 12
```

Error: no Snakefile found, tried Snakefile, snakefile, workflow/Snakefile, workflow/snakefile.

"Snakefile" is one of the default filenames that snakemake automatically looks for when looking for rules.

We can explicitly specify the Snakefile to read by using `-s` as shown in this fake example

```
snakemake -s MyFirstSnakefile1.snake <targets>
```



Our snakefile contains two rules that tell `snakemake` how to make plots and filter the files.

```
cat Snakefile1
```

```
rule plot:
    output: "{dataset}_plot.pdf"
    input:  "{dataset}.csv"
    shell:  "myplotter.py -o {output} {input}"

rule filter:
    output: "{csvdata}_filtered.csv"
    input:  "{csvdata}.csv"
    shell:  "egrep -v ^boring {input} > {output}"
```

Indentation etc

As you have probably noticed....like Python, snakemake uses line indentation to indicate the hierarchy of the lines in the Snakefile.

The `rule:` lines have no spaces in front of them.

The `output:` `input:` and `shell:` lines have four spaces in front of them, and the actual shell commands have eight spaces.

The actual number is unimportant, so long as everything lines up.

Let's try it!

ls

ds1.csv ds2.csv myplotter.py Snakefile1 ...

The **-pn** flag tells snakemake two things. **p** gets it to display the actual shell commands as it runs them, which is almost always a good idea, and **n** means just do a dry-run, without actually doing it all (an excellent idea always!)

Run it with -p and -n

snakemake -pn

Error: no Snakefile found, tried Snakefile, snakefile, workflow/Snakefile, workflow/snakefile..

Run it with -p and -n AND tell it which file to read...

snakemake -pn -s Snakefile1

Building DAG of jobs...

WorkflowError:

Target rules may not contain wildcards.

Please specify concrete files or a rule without wildcards at the command line, or have a rule without wildcards at the very top of your workflow (e.g. the typical "rule all" which just collects all results you want to generate in the end).

What happens if we try this command, which

specifies the output target ds1_plot.pdf

snakemake -pn ds1_plot.pdf -s Snakefile1

Building DAG of jobs...

Job stats:

job	count	min threads	max threads
plot	1	1	1
total	1	1	1

[Thurs Nov 27 09:37:06 2025]

rule plot:

input: ds1.csv

output: ds1_plot.pdf

jobid: 0

reason: Missing output files: ds1_plot.pdf

wildcards: dataset=ds1

resources: tmpdir=/tmp

./myplotter.py -o ds1_plot.pdf ds1.csv

Job stats:

job	count	min threads	max threads
plot	1	1	1
total	1	1	1

Reasons:

(check individual jobs above for details)

output files have to be generated:

plot

This was a dry-run (flag -n). The order of jobs does not reflect the order of execution.

[Give feedback](#)

[Ask AI for help](#)

When the `snakemake` command is run in the shell, it reads the Snakefile and looks for rules that will produce the requested output. It fills in the wildcards in the rule definitions to make a job.

What rules can be met?

Our command was

```
snakemake -pn ds1_plot.pdf -s Snakefile1
```

The first rule in the Snakefile1 file was

```
rule plot:
    output: "{dataset}_plot.pdf"
    input:  "{dataset}.csv"
    shell:  " ./myplotter.py -o {output} {input}"
```

`{dataset}` is thus inferred by snakemake to be `ds1` because we asked for `ds1_plot.pdf`

Thus if the `input:` exists, the `output:` bit can be done, as long as the `shell:` bit is correct/possible

What do you think will happen when the command we give is

```
snakemake -pn ds1_plot.pdf ds2_filtered_plot.pdf -s Snakefile1
```

We can predict that snakemake will run three separate jobs (one filter job and two plot jobs) to make the desired outputs. **HOW?!**

Let's look at our Snakefile1 again

```
rule plot:
    output: "{dataset}_plot.pdf"
    input:  "{dataset}.csv"
    shell:  " ./myplotter.py -o {output} {input}"

rule filter:
    output: "{csvdata}_filtered.csv"
    input:  "{csvdata}.csv"
    shell:  "egrep -v ^boring {input} > {output}"
```

Question: What do we want at the end?

Answer: *ds2_filtered_plot.pdf*

Question: How do we get there?

Answer: *do the filtering, then do the plot*

Question: What do we need to do the plot?

Answer: *filtered csv data*

Question: How do we get the filtered csv?

Answer: *take the raw data and filter it*

Question: Which raw data?

Answer: *ds2.csv, if it is there*

Looking at the filter rule `{csvdata}` "could" be ds1 and/or ds2, but we only wanted a filtered plot of the ds2 data, so `{csvdata}` would be inferred to be ds2, thus:

- input will be `ds2.csv`
- output will be `ds2_filtered.csv`
- process will be `egrep -v ^boring ds2.csv > ds2_filtered.csv`

As we have specified the target (`output`) as `ds2_filtered_plot.pdf`, and `ds2_filtered.csv` now satisfies the `input` for rule `plot`: we can do the plot using the `shell` component

The output plot is called `ds2_filtered_plot.pdf`, so `{dataset}` for this one will be "ds2_filtered"

We also said we wanted `ds1_plot.pdf` as an output, not filtered. `{dataset}` for this one will be "ds1", so if `ds1.csv` exists as a possible input, the rule can be applied (it does, so it should be!)

Let's see what a dry run tells us

```
snakemake -pn ds1_plot.pdf ds2_filtered_plot.pdf -s Snakefile1
```

```
Building DAG of jobs...
```

```
Job stats:
```

job	count	min threads	max threads
filter	1	1	1
plot	2	1	1
total	3	1	1

```
[Thurs Nov 27 09:38:59 2025]
```

```
rule filter:
```

```
  input: ds2.csv
```

```
  output: ds2_filtered.csv
```

```
  jobid: 1
```

```
  reason: Missing output files: ds2_filtered.csv
```

```
  wildcards: csvdata=ds2
```

```
  resources: tmpdir=/tmp
```

```
egrep -v ^boring ds2.csv > ds2_filtered.csv
```

```
[Thurs Nov 27 09:38:59 2025]
```

```
rule plot:
```

```
  input: ds1.csv
```

```
  output: ds1_plot.pdf
```

```
  jobid: 2
```

```
  reason: Missing output files: ds1_plot.pdf
```

```
  wildcards: dataset=ds1
```

```
  resources: tmpdir=/tmp
```

```
./myplotter.py -o ds1_plot.pdf ds1.csv
```

```
[Thurs Nov 27 09:38:59 2025]
```

```
rule plot:
```

```
  input: ds2_filtered.csv
```

```
  output: ds2_filtered_plot.pdf
```

```
  jobid: 0
```

```
  reason: Missing output files: ds2_filtered_plot.pdf; Input files updated by another job
```

```
  wildcards: dataset=ds2_filtered
```

```
  resources: tmpdir=/tmp
```

```
./myplotter.py -o ds2_filtered_plot.pdf ds2_filtered.csv
```

Job stats:

job	count	min threads	max threads
filter	1	1	1
plot	2	1	1
total	3	1	1

Reasons:

- (check individual jobs above for details)
- input files updated by another job:
 - plot
- output files have to be generated:
 - filter, plot

This was a dry-run (flag -n). The order of jobs does not reflect the order of execution.

We were correct!

Let's do it for real! Use `-p` instead of `-pn`

```
snakemake -p ds1_plot.pdf ds2_filtered_plot.pdf -s Snakefile1
```

Assuming unrestricted shared filesystem usage.

Error: cores have to be specified for local execution (use --cores N with N being a number

Let's try again, with `--cores 1`

```
snakemake -p --cores 1 ds1_plot.pdf ds2_filtered_plot.pdf -s Snakefile1
```

Building DAG of jobs...

Using shell: /usr/bin/bash

Provided cores: 1 (use --cores to define parallelism)

Rules claiming more threads will be scaled down.

Job stats:

job	count	min threads	max threads
filter	1	1	1
plot	2	1	1
total	3	1	1

Select jobs to execute...

[Thurs Nov 27 09:40:56 2025]

rule filter:

input: ds2.csv

output: ds2_filtered.csv

jobid: 1

reason: Missing output files: ds2_filtered.csv

wildcards: csvdata=ds2

resources: tmpdir=/tmp

```
egrep -v ^boring ds2.csv > ds2_filtered.csv
```

[Thurs Nov 27 09:40:56 2025]

Finished job 1.

1 of 3 steps (33%) done

Select jobs to execute...

[Thurs Nov 27 09:40:56 2025]

rule plot:

input: ds1.csv

output: ds1_plot.pdf

jobid: 2

reason: Missing output files: ds1_plot.pdf

wildcards: dataset=ds1

resources: tmpdir=/tmp


```
./myplotter.py -o ds1_plot.pdf ds1.csv
[Thurs Nov 27 09:40:58 2025]
Finished job 2.
2 of 3 steps (67%) done
Select jobs to execute...

[Thurs Nov 27 09:40:58 2025]
rule plot:
  input: ds2_filtered.csv
  output: ds2_filtered_plot.pdf
  jobid: 0
  reason: Missing output files: ds2_filtered_plot.pdf; Input files updated by another job
  wildcards: dataset=ds2_filtered
  resources: tmpdir=/tmp

./myplotter.py -o ds2_filtered_plot.pdf ds2_filtered.csv
[Thurs Nov 27 09:40:59 2025]
Finished job 0.
3 of 3 steps (100%) done
Complete log: .snakemake/log/2024-11-27T094056.632825.snakemake.log
```

Let's add a bit more processing power to our work, passing the `--cores 12` parameter to snakemake

```
snakemake -p --cores 12 ds1_plot.pdf ds2_filtered_plot.pdf -s Snakefile1
```

```
Assuming unrestricted shared filesystem usage.
host: bioinfmsc8.bio.ed.ac.uk
Building DAG of jobs...
Nothing to be done (all requested files are present and up to date).
```

HUH?

Why? Because all targets are present: the job is obviously complete!

Remove one of the targets and re-run

```
rm ds2_filtered_plot.pdf
```

```
snakemake -p --cores 12 ds1_plot.pdf ds2_filtered_plot.pdf -s Snakefile1
```

```
Building DAG of jobs...
Using shell: /usr/bin/bash
Provided cores: 12
Rules claiming more threads will be scaled down.
Job stats:
job      count      min threads      max threads
-----  -
plot          1              1              1
total         1              1              1

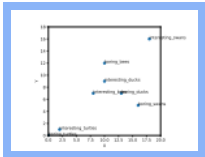
Select jobs to execute...

[Thurs Nov 27 09:42:31 2025]
rule plot:
  input: ds2_filtered.csv
  output: ds2_filtered_plot.pdf
  jobid: 1
  reason: Missing output files: ds2_filtered_plot.pdf
  wildcards: dataset=ds2_filtered
  resources: tmpdir=/tmp

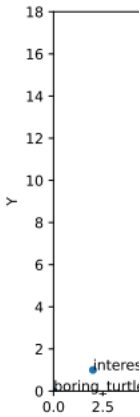
./myplotter.py -o ds2_filtered_plot.pdf ds2_filtered.csv
[Thurs Nov 27 09:42:33 2025]
Finished job 1.
1 of 1 steps (100%) done
Complete log: .snakemake/log/2024-11-27T094231.419141.snakemake.log
```

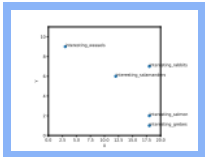
Give feedback
Ask AI for help

Let's have a look at our PDFs!
evince *.pdf &

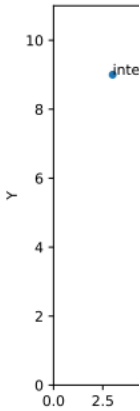


1





1



What would you predict would happen when the command we give is

snakemake --cores 12 -p ds3_filtered_plot.pdf -s Snakefile1

BAssuming unrestricted shared filesystem usage.

host: bioinfmsc8.bio.ed.ac.uk

Building DAG of jobs...

MissingInputException in rule filter in file "/localdisk/home/aivens2/Exercises/Lecture22/

Missing input files for rule filter:

output: ds3_filtered.csv

wildcards: csvdata=ds3

affected files:

ds3.csv

Logic is based on:

- file name patterns
- {wildcard} placeholders
- snakemake starts with a target filename and looks for rules with a matching output
- repeats until it runs out of rules
- order of rules in the snakefile is NOT IMPORTANT!

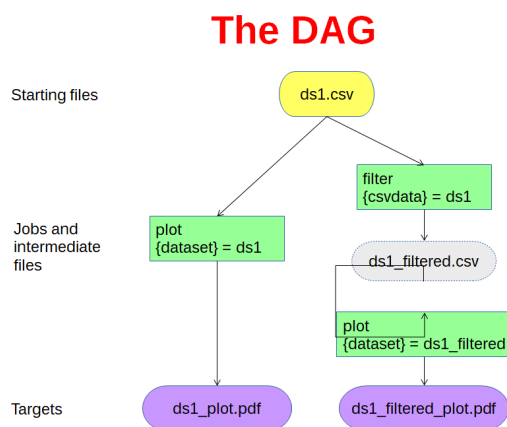
All the `shell:` commands run in subshells, so if we change or create environment variables, change current working directory, it does not affect other rules: rules always communicate via files.

📌What are these things called DAGs?

The set of jobs that snakemake works out will always form what is called a [directed acyclic graph](#) or DAG.

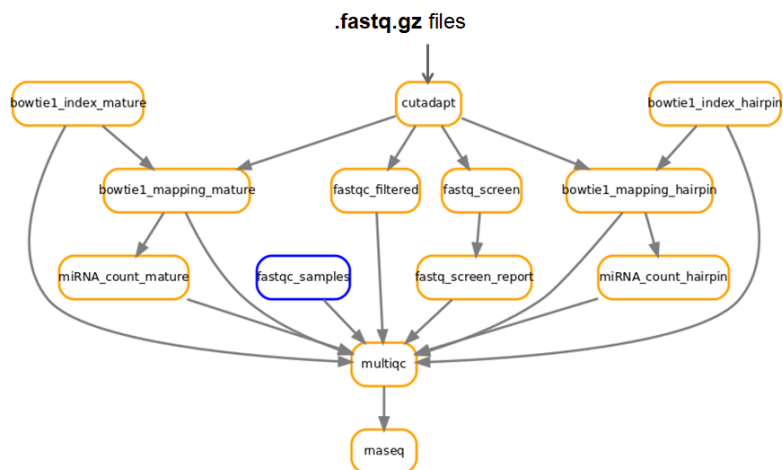
The jobs in the DAG can now be executed in any order so long as all the prior jobs have completed.

Here's one that could have been derived from our Snakefile1



Hmm... memories of MyFirstPipeline...!

DAGs can get complex



https://sequana.readthedocs.io/en/master/pipeline_smallnaseq.html

⬆Inputs and outputs can be named

WAS

```
rule plot:
    output: "{dataset}_plot.pdf"
    input:  "{dataset}.csv"
    shell:  "./myplotter.py -o {output} {input}"

rule filter:
    output: "{csvdata}_filtered.csv"
    input:  "{csvdata}.csv"
    shell:  "egrep -v ^boring {input} > {output}"
```

UPDATED version `Snakefile1_named`

```
rule plot:
    output:
        plot = "{dataset}_plot.pdf"
    input:
        csv = "{dataset}.csv"
    shell:
        "myplotter.py -o {output.plot} {input.csv}"

rule filter:
    output:
        csv = "{csvdata}_filtered.csv"
    input:
        csv = "{csvdata}.csv"
    shell:
        "egrep -v ^boring {input.csv} > {output.csv}"
```

If a rule has multiple distinct inputs or outputs, that's absolutely fine, just give them names!

The shell: command must be in quotes

```
shell:
    "wc {input} > {output}"
```

For multiple commands, use triple-quotes and break long

lines with a backslash (the following is just example syntax, not a real thing!)

```
shell:
    r"""make_index {input}
        /path/to/my/bin/long_command.py \
        --input_file={input}.idx \
        --consensus_out={output.consensus}
    """
```

📄 Variables and Functions

As well as rules, we can define variables (or rather, they are actually constants, as snakemake rules cannot modify them!) in our Snakefile.

```
# A simple (string) variable
```

```
adapter = "GAGTAGTAT"
```

```
# A list variable
```

```
ranks = [ "kingdom", "phylum", "genus" ]
```

```
# Combining variables...
```

Lists are often useful in snakemake workflows. You can write them out explicitly as normal Python lists, or generate them from a template with the `expand(...)` function:

```
ranks = [ "kingdom", "phylum", "genus" ]
```

```
modi = [ "super", "sub", "" ]
```

```
expand("{m}{r}_list.txt", r=ranks, m=modi)
```

```
# The expand line does all the combinations of the pattern, to generate a list
```

```
["superkingdom_list.txt", "superphylum_list.txt", "supergenuss_list.txt", "subkingdom_list.txt",  
"subphylum_list.txt", "subgenuss_list.txt", "kingdom_list.txt", "phylum_list.txt", "genuss_list.txt"]
```

```
# Shell command outputs to make lists : similar to using glob
```

```
import glob
```

```
glob.glob("*.csv")
```

```
['file1.csv', 'file2.csv']
```

```
# shell command approach example
```

```
# using sed to edit on the fly
```

```
list(shell( "ls *.csv | sed 's/.csv/.pdf/'", iterable = True ))
```

```
# Would result in this going to snakemake
```

```
[ 'file1.pdf', 'file2.pdf' ]
```

📌 Things in { curly brackets }

In snakemake, you see a lot of things in curly brackets. Each is a placeholder, but the context is important.

- In the input: and output: strings they are wildcards, and must match part of the filename.
- In the shell: part of shell() function they are placeholders which may be:
- {input} - The input file(s)
- {output} - The output file(s)
- {wildcards.dataset} - The value of the "dataset" wildcard
- {adapter} - The "adapter" variable defined at the top of the Snakefile
- Within the expand() function they are placeholders to be expanded, and must be mapped to lists of values after the template
- In all cases, if you need to preserve the literal brackets, you need to double them up, eg. when using awk within a shell command:

```
shell:  
    "awk '{{print $1}}' {input} > {output}"
```

Plot all {data}/*.csv files in one combined plot

Use of "lambda" defers calculating the list

Here are the contents of a new snakefile I have called foosnake

```
rule plot:
    output:
        "{data}_plot.pdf"
    input:
        lambda x: shell("ls {x.data}/*.csv", iterable=True)
    shell:
        "./myplotter.py -o {output} {input}"
```

EXAMPLE: Say we have files in a directory called dfiles

ls ./dfiles/*

./dfiles/ds1.csv ./dfiles/ds2.csv ./dfiles/ds2_filtered.csv

ls dfiles

ds1.csv ds2.csv ds2_filtered.csv

EXAMPLE only

snakemake --cores 1 dfiles_plot.pdf -s foosnake

Building DAG of jobs...

Using shell: /usr/bin/bash

Provided cores: 1 (use --cores to define parallelism)

Rules claiming more threads will be scaled down.

Job stats:

job	count	min threads	max threads
plot	1	1	1
total	1	1	1

Select jobs to execute...

[Thurs Nov 27 10:31:47 2025]

rule plot:

input: dfiles/ds1.csv, dfiles/ds2.csv, dfiles/ds2_filtered.csv

output: dfiles_plot.pdf

jobid: 0

reason: Missing output files: dfiles_plot.pdf

wildcards: data=dfiles

resources: tmpdir=/tmp

[Thurs Nov 27 10:31:49 2025]

Finished job 0.

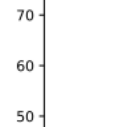
1 of 1 steps (100%) done

Complete log: .snakemake/log/2024-11-27T132842.826526.snakemake.log

EXAMPLE only

evince dfiles_plot.pdf &

[1] 2369483



EXAMPLE only: If we have a folder called MyDir1 which has files that we want to process...

Is MyDir1/

indir1.csv indir2.csv indir2_filtered.csv

EXAMPLE only: Alternative syntax with a named function

Here is our snakefile called `function_snake`

```
def plot_input(x):
    return shell("ls {x.data}/*.csv", iterable=True)

rule plot:
    output:
        "{data}_plot.pdf"
    input:
        plot_input
    shell:
        "./myplotter.py -o {output} {input}"
```

EXAMPLE only: Let's do a dry run to produce the pdf for the MyDir1 files

snakemake --cores 1 -pn MyDir1_plot.pdf -s function_snake

Building DAG of jobs...

ls MyDir1/*.csv

Job stats:

job	count	min threads	max threads
plot	1	1	1
total	1	1	1

[Thurs Nov 27 10:40:52 2025]

rule plot:

input: MyDir1/indir1.csv, MyDir1/indir2.csv, MyDir1/indir2_filtered.csv

output: MyDir1_plot.pdf

jobid: 0

reason: Missing output files: MyDir1_plot.pdf

{data} becomes MyDir1, iterated by the function to produce the list of files for the

wildcards: data=MyDir1

resources: tmpdir=/tmp

`./myplotter.py -o MyDir1_plot.pdf MyDir1/indir1.csv MyDir1/indir2.csv MyDir1/indir2_filter`

Job stats:

job	count	min threads	max threads
plot	1	1	1
total	1	1	1

Reasons:

(check individual jobs above for details)

output files have to be generated:

plot

This was a dry-run (flag -n). The order of jobs does not reflect the order of execution.

EXAMPLE only: Run it for real....

snakemake --cores 1 -p MyDir1_plot.pdf -s function_snake

Building DAG of jobs...

ls MyDir1/*.csv

Using shell: /usr/bin/bash

Provided cores: 1 (use --cores to define parallelism)

```
Rules claiming more threads will be scaled down.
Job stats:
job      count      min threads      max threads
-----
plot      1              1              1
total     1              1              1

Select jobs to execute...

[Thurs Nov 27 10:43:38 2025]
rule plot:
  input: MyDir1/indir1.csv, MyDir1/indir2.csv, MyDir1/indir2_filtered.csv
  output: MyDir1_plot.pdf
  jobid: 0
  reason: Missing output files: MyDir1_plot.pdf
  wildcards: data=MyDir1
  resources: tmpdir=/tmp

./myplotter.py -o MyDir1_plot.pdf MyDir1/indir1.csv MyDir1/indir2.csv MyDir1/indir2_filter
[Thurs Nov 27 10:43:39 2025]
Finished job 0.
1 of 1 steps (100%) done
Complete log: .snakemake/log/2024-11-27T104337.786769.snakemake.log
```

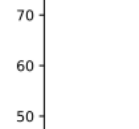
EXAMPLE only: Did it work?

Is -lart | tail -4

drwxr-xr-x 1 aivens2 g_aivens2 180 Nov 27 10:39 MyDir1
-rw-r--r-- 1 aivens2 g_aivens2 187 Nov 27 10:40 function_snake
drwxr-xr-x 1 aivens2 g_aivens2 524 Nov 27 10:43 .
-rw-r--r-- 1 aivens2 g_aivens2 16190 Nov 27 10:43 MyDir1_plot.pdf

View the PDF

evince MyDir1_plot.pdf &



...and finally...

Silly filenames: try NOT to use unusual file names, e.g. "We\$tward Ho!.txt" or "John O'Groats.txt"

This fix won't always work if either have whitespace in them

```
shell: "wc '{input}' > '{output}'"
```

This is robust: `:q` lets Snakemake quote each of the elements that contains whitespace

```
shell: "wc {input:q} > {output:q}"
```

A lot of the syntax we're using in snakemake is inherited from Python, so we can use it in snakemake

Rules can run Python code using `run:` instead of `shell:`

```
shell:
    """for i in {input} ; do
        sort -u $i >> {output}
    done
    """

run:
    with open(output[0], "w") as myConnection:
        for i in input:
            with open(i) as someInputs:
                myConnection.write(sorted(set(someInputs)))
```


📌 A snakemake workflow

We will use snakemake to create a simple workflow to assemble short reads (courtesy of [Mark Blaxter!](#)) from a bacterial genome into contigs using a [De Bruijn graph](#) assembler called [velvet](#) (this used to give an amusing 404 error, but now it is a rather staid 500 error).

Ha ha! It's all about kmers again....!



[Daniel Zerbino's PhD thesis](#)

The velvet DNA sequence assembler IS already installed on the server for you ([see how!](#))

```
# Make it so that the assembly processes can be done in parallel
```

```
# https://www.openmp.org/
```

```
OMP_THREAD_LIMIT=24
```

```
# Help?!
```

```
velveth -h
```

```
velveth - simple hashing program  
Version 1.2.10
```

```
Copyright 2007, 2008 Daniel Zerbino (zerbino@ebi.ac.uk)
```

```
This is free software; see the source for copying conditions. There is NO  
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

Compilation settings:

CATEGORIES = 2

MAXKMERLENGTH = 31

Usage:

```
./velveth directory hash_length [[-file_format][-read_type][-separate|-interleaved] filename
```

```
directory      : directory name for output files
hash_length    : EITHER an odd integer (if even, it will be decremented) <= 31 (i
                 : OR: m,M,s where m and M are odd integers (if not, they will be d
                 : and s is a step (even number). velvet will then hash from
filename       : path to sequence file or - for standard input
```

File format options:

```
-fasta -fastq -raw -fasta.gz -fastq.gz -raw.gz -sam -bam -f
(Note: -fmtAuto will detect fasta or fastq, and will try the following programs fo
```

File layout options for paired reads (only for fasta and fastq formats):

```
-interleaved : File contains paired reads interleaved in the one file (default)
-separate    : Read 2 separate files for paired reads
```

Read type options:

```
-short -shortPaired
-short2 -shortPaired2
-long -longPaired
-reference
```

Options:

```
-strand_specific : for strand specific transcriptome sequencing data (defau
-reuse_Sequences : reuse Sequences file (or link) already in directory (no
-reuse_binary    : reuse binary sequences file (or link) already in directory (no n
-noHash          : simply prepare Sequences file, do not hash reads or prep
-create_binary   : create binary CnyUnifiedSeq file (default: off)
```

Synopsis:

- Short single end reads:
velveth Assem 29 -short -fastq s_1_sequence.txt
- Paired-end short reads (remember to interleave paired reads):
velveth Assem 31 -shortPaired -fasta interleaved.fna
- Paired-end short reads (using separate files for the paired reads)
velveth Assem 31 -shortPaired -fasta -separate left.fa right.fa
- Two channels and some long reads:
velveth Assem 43 -short -fastq unmapped.fna -longPaired -fasta SangerReads.fasta
- Three channels:
velveth Assem 35 -shortPaired -fasta pe_lib1.fasta -shortPaired2 pe_lib2.fasta -sh

Output:

```
directory/Roadmaps
directory/Sequences
[Both files are picked up by graph, so please leave them there]
```

🏠Phew!

To assess assembly quality, we will calculate the N50 using the "stats.sh" script within the [BBMap aligner package](#).

N50 is a statistic that describes the distribution of contig lengths and estimates the completeness of the assembly.

We could add in a step prior to velvet assembly to use a quality and adapter read trimming software called [cutadapt](#), and check how this affects the assembly quality, but today, we are not going to as there is really NOT enough time... a Python module is installed... just try this in a terminal window...

cutadapt

```
This is cutadapt 4.4 with Python 3.12.3
See https://cutadapt.readthedocs.io/ for full documentation.
Command line parameters:
ERROR: You did not provide any input file names. Please give me something to do!
```

We will use a parameter in the velvet assembler to vary the kmer size.

Kmer size is the length of sequences used in creating sequence connections during assembly, and optimal kmer size is dependent on the read length and depth and complexity of sequence.

While many other parameters and statistics are available, for simplicity today we will simply tune the Kmer size and use the N50 statistic to choose the best assembly.

velvet itself is always run in two steps

1. `velveth` produces a hash table which is used by velvetg in the next step
2. `velvetg` builds and manipulates the De Bruijn graph and produces the assembly

We will build up to a full snakemake workflow in stages

1. Running the velvet and stats.sh commands manually
2. Running the velvet commands via a snakemake rule
3. Adding a second rule to run stats.sh
4. Enable our workflow to use a range of kmer sizes to find the optimum kmer size

📌Running a velvet assembly manually

```
ls -l /localdisk/data/BPSM/Lecture22/*.gz
```

```
/localdisk/data/BPSM/Lecture22/6991_1.fastq.gz  
/localdisk/data/BPSM/Lecture22/6991_2.fastq.gz  
/localdisk/data/BPSM/Lecture22/A_Tco_1.fastq.gz  
/localdisk/data/BPSM/Lecture22/A_Tco_2.fastq.gz
```

There are two sets of paired end reads in `/localdisk/data/BPSM/Lecture22/` called

- `6991_1.fastq.gz` and `6991_2.fastq.gz`: these are reads from a bacterial genome project
- `A_Tco_1.fastq.gz` and `A_Tco_2.fastq.gz`: these are some of the reads from the *Trypanosoma congolense* RNAseq analysis we did for ICA1...

We will try the bacterial one first...

```
# We run velveth first (easiest using bash variables here...)
```

```
fq1="/localdisk/data/BPSM/Lecture22/6991_1.fastq.gz"  
fq2="/localdisk/data/BPSM/Lecture22/6991_2.fastq.gz"  
velveth paired_k21 21 -shortPaired -fastq.gz -separate ${fq1} ${fq2}  
  
[0.000000] Reading FastQ file /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz;  
[0.000999] Reading FastQ file /localdisk/data/BPSM/Lecture22/6991_2.fastq.gz;  
[5.564363] 3846320 sequences found in total in the paired sequence files  
[5.564391] Done  
[8.818598] Reading read set file paired_k21/Sequences;  
[9.113403] 3846320 sequences found  
[11.010423] Done  
[11.010454] 3846320 sequences in total.  
[11.012081] Writing into roadmap file paired_k21/Roadmaps...  
[12.702968] Inputting sequences...  
[12.703021] Inputting sequence 0 / 3846320  
[18.215648] Inputting sequence 1000000 / 3846320  
[24.261270] Inputting sequence 2000000 / 3846320  
[30.343522] Inputting sequence 3000000 / 3846320  
[35.564465] === Sequences loaded in 22.899760 s  
[37.421241] Done inputting sequences  
[37.421274] Destroying splay table  
[37.469097] Splay table destroyed
```

```
# What files did we get?
```

```
ls -alh paired_k21
```

```
total 499M  
drwxr-xr-x 1 aivens2 users 128 Nov 27 13:49 .  
drwxr-xr-x 1 aivens2 864524 544 Nov 27 13:49 ..  
-rw-r--r-- 1 aivens2 users 454 Nov 27 13:49 Log  
-rw-r--r-- 1 aivens2 users 193M Nov 27 13:49 Roadmaps  
-rw-r--r-- 1 aivens2 users 349M Nov 27 13:49 Sequences
```

```
# Now we run velvetg, which uses data from velveth
```

```
velvetg paired_k21
```

```
[0.000000] Reading roadmap file paired_k21/Roadmaps
[4.646243] 3846320 roadmaps read
[4.649609] Creating insertion markers
[5.353829] Ordering insertion markers
[7.629289] Counting preNodes
[8.137945] 3697105 preNodes counted, creating them now
[9.953407] Sequence 1000000 / 3846320
[10.890965] Sequence 2000000 / 3846320
[11.800134] Sequence 3000000 / 3846320
[12.552074] Adjusting marker info...
[13.119388] Connecting preNodes
[13.804541] Connecting 1000000 / 3846320
[14.617893] Connecting 2000000 / 3846320
[15.436414] Connecting 3000000 / 3846320
[16.132680] Cleaning up memory
[16.148890] Done creating preGraph
[16.148924] Concatenation...
[16.824307] Renumbering preNodes
[16.824349] Initial preNode count 3697105
[16.867465] Destroyed 3447860 preNodes
[16.867493] Concatenation over!
[16.867497] Clipping short tips off preGraph
[16.918718] Concatenation...
[17.136633] Renumbering preNodes
[17.136670] Initial preNode count 249245
[17.146245] Destroyed 150714 preNodes
[17.146277] Concatenation over!
[17.146281] 79749 tips cut off
[17.146286] 98531 nodes left
[17.147883] Writing into pregraph file paired_k21/PreGraph...
[17.606692] Reading read set file paired_k21/Sequences;
[17.882810] 3846320 sequences found
[19.720597] Done
[21.433398] Reading pre-graph file paired_k21/PreGraph
[21.433537] Graph has 98531 nodes and 3846320 sequences
[21.598105] Scanning pre-graph file paired_k21/PreGraph for k-mers
[21.646618] 6365098 kmers found
[21.847067] Sorting kmer occurrence table ...
[23.257595] Sorting done.
[23.257634] Computing acceleration table...
[23.338718] Computing offsets...
[23.354924] Ghost Threading through reads 0 / 3846320
[29.358156] Ghost Threading through reads 1000000 / 3846320
[35.388199] Ghost Threading through reads 2000000 / 3846320
[41.336156] Ghost Threading through reads 3000000 / 3846320
[46.369954] === Ghost-Threaded in 23.015031 s
[46.370005] Threading through reads 0 / 3846320
[53.256560] Threading through reads 1000000 / 3846320
[60.143886] Threading through reads 2000000 / 3846320
[66.986634] Threading through reads 3000000 / 3846320
[72.748053] === Threaded in 26.378048 s
[72.760326] Correcting graph with cutoff 0.200000
[72.763213] Determining eligible starting points
[72.844962] Done listing starting nodes
[72.844999] Initializing todo lists
[72.862636] Done with initialization
[72.862673] Activating arc lookup table
[72.877746] Done activating arc lookup table
[72.928569] 10000 / 98531 nodes visited
[72.981956] 20000 / 98531 nodes visited
[73.040891] 30000 / 98531 nodes visited
[73.100620] 40000 / 98531 nodes visited
[73.169025] 50000 / 98531 nodes visited
[73.235830] 60000 / 98531 nodes visited
[73.305673] 70000 / 98531 nodes visited
[73.382346] 80000 / 98531 nodes visited
[73.448291] 90000 / 98531 nodes visited
[73.480933] 100000 / 98531 nodes visited
[73.485924] Concatenation...
```

```
[73.489732] Renumbering nodes
[73.489754] Initial node count 98531
[73.490691] Removed 66587 null nodes
[73.490709] Concatenation over!
[73.490714] Clipping short tips off graph, drastic
[73.499111] Concatenation...
[73.516539] Renumbering nodes
[73.516576] Initial node count 31944
[73.517273] Removed 7695 null nodes
[73.517285] Concatenation over!
[73.517289] 24249 nodes left
[73.519444] Writing into graph file paired_k21/Graph...
[74.118746] WARNING: NO COVERAGE CUTOFF PROVIDED
[74.118785] Velvet will probably leave behind many detectable errors
[74.118797] See manual for instructions on how to set the coverage cutoff parameter
[74.118823] Removing contigs with coverage < -1.000000...
[74.119960] Concatenation...
[74.122803] Renumbering nodes
[74.122820] Initial node count 24249
[74.122956] Removed 0 null nodes
[74.123078] Concatenation over!
[74.123618] Concatenation...
[74.125077] Renumbering nodes
[74.125089] Initial node count 24249
[74.125112] Removed 0 null nodes
[74.125209] Concatenation over!
[74.125228] Clipping short tips off graph, drastic
[74.125711] Concatenation...
[74.126825] Renumbering nodes
[74.126837] Initial node count 24249
[74.126947] Removed 0 null nodes
[74.126953] Concatenation over!
[74.127058] 24249 nodes left
[74.127063] WARNING: NO EXPECTED COVERAGE PROVIDED
[74.127067] Velvet will be unable to resolve any repeats
[74.127070] See manual for instructions on how to set the expected coverage parameter
[74.127165] Concatenation...
[74.128279] Renumbering nodes
[74.128292] Initial node count 24249
[74.128315] Removed 0 null nodes
[74.128410] Concatenation over!
[74.128419] Removing reference contigs with coverage < -1.000000...
[74.128737] Concatenation...
[74.129845] Renumbering nodes
[74.129857] Initial node count 24249
[74.129966] Removed 0 null nodes
[74.130068] Concatenation over!
[74.142753] Writing contigs into paired_k21/contigs.fa...
[74.529659] Writing into stats file paired_k21/stats.txt...
[74.595746] Writing into graph file paired_k21/LastGraph...
Final graph has 24249 nodes and n50 of 1524, max 10406, total 5800362, using 0/3846320 res
```

```
# Now we need to obtain information on our assembly, and
# find the number of contigs and the contig N50. This uses
# a command called stats.sh from the BBMap project,
# which is already installed.
```

```
# Let's get the stats (see how!)!
```

```
stats.sh paired_k21/contigs.fa
```

A	C	G	T	N	IUPAC	Other	GC	GC_stddev
0.2502	0.2497	0.2516	0.2485	0.0000	0.0000	0.0000	0.5013	0.0841
Main genome scaffold total:					16964			
Main genome contig total:					16964			
Main genome scaffold sequence total:					6.084 MB			
Main genome contig sequence total:					6.084 MB			
							0.000% gap	

Main genome scaffold N/L50: 1126/1.447 KB
Main genome contig N/L50: 1126/1.447 KB
Main genome scaffold N/L90: 7691/99
Main genome contig N/L90: 7691/99
Max scaffold length: 10.426 KB
Max contig length: 10.426 KB
Number of scaffolds > 50 KB: 0
% main genome in scaffolds > 50 KB: 0.00%

Minimum Scaffold Length	Number of Scaffolds	Number of Contigs	Total Scaffold Length	Total Contig Length	Scaffold Contig Coverage
All	16,964	16,964	6,084,365	6,084,365	100.00%
50	14,849	14,849	5,994,624	5,994,624	100.00%
100	7,634	7,634	5,471,737	5,471,737	100.00%
250	4,070	4,070	4,946,515	4,946,515	100.00%
500	2,786	2,786	4,491,794	4,491,794	100.00%
1 KB	1,646	1,646	3,675,722	3,675,722	100.00%
2.5 KB	480	480	1,819,951	1,819,951	100.00%
5 KB	69	69	441,619	441,619	100.00%
10 KB	4	4	40,965	40,965	100.00%

🏠 Running a velvet assembly in snakemake

In order to capture the general process, we will put these commands into a snakemake rule file; let's call it

`velvet_snakefile1`.

Note how we do NOT need to copy the raw data files!

```
rule assemble:
    output: "{sample}_paired_k21_contigs.fa"
    input:
        read1 = "/localdisk/data/BPSM/Lecture22/{sample}_1.fastq.gz",
        read2 = "/localdisk/data/BPSM/Lecture22/{sample}_2.fastq.gz"
    shell:
        r"""velveth paired_k21 21 -shortPaired -fastq -separate {input.read1} {input.read2}
        velvetg paired_k21
        mv paired_k21/contigs.fa {output}
        """
```

Some things to note about this snakemake rule

- We have chosen to run both stages of velvet as a single shell job
- Triple-double quotes (r"") are used to enclose multiple shell commands
- The {sample} wildcard is used in all input and output filenames
- There are two input files, so we give them names
- In the shell part, the named inputs are referenced using curly brackets and a dot after input or output, e.g. {input.read1}
- We use a mv command at the end to rename the output file, because velvetg does not allow us to specify the output filename directly
- We're ignoring the various other files that velvet produces

Does it work?! "Dry run" version first

snakemake -pn 6991_paired_k21_contigs.fa -s velvet_snakefile1

Building DAG of jobs...

Job stats:

job	count	min threads	max threads
assemble	1	1	1
total	1	1	1

[Thurs Nov 27 14:24:30 2025]

rule assemble:

```
input: /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz, /localdisk/data/BPSM/Lecture22/
output: 6991_paired_k21_contigs.fa
jobid: 0
reason: Missing output files: 6991_paired_k21_contigs.fa
wildcards: sample=6991
resources: tmpdir=/tmp
```

```
velveth paired_k21 21 -shortPaired -fastq -separate /localdisk/data/BPSM/Lecture22/6991_1.
velvetg paired_k21
```



```
mv paired_k21/contigs.fa 6991_paired_k21_contigs.fa
```

Job stats:

job	count	min threads	max threads
assemble	1	1	1
total	1	1	1

Reasons:

(check individual jobs above for details)
output files have to be generated:
assemble

This was a dry-run (flag -n). The order of jobs does not reflect the order of execution.

OK, looks like the snakemake part was OK.

Let's do it for real, just take the "n" off the -pn

```
snakemake -p --cores 1 6991_paired_k21_contigs.fa -s velvet_snakefile1
```

Building DAG of jobs...

Using shell: /usr/bin/bash

Provided cores: 1 (use --cores to define parallelism)

Rules claiming more threads will be scaled down.

Job stats:

job	count	min threads	max threads
assemble	1	1	1
total	1	1	1

Select jobs to execute...

[Thurs Nov 27 14:25:48 2025]

rule assemble:

input: /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz, /localdisk/data/BPSM/Lecture22/
output: 6991_paired_k21_contigs.fa
jobid: 0
reason: Missing output files: 6991_paired_k21_contigs.fa
wildcards: sample=6991
resources: tmpdir=/tmp

```
velveth paired_k21 21 -shortPaired -fastq -separate /localdisk/data/BPSM/Lecture22/6991_1.  
velvetg paired_k21  
mv paired_k21/contigs.fa 6991_paired_k21_contigs.fa
```

```
[0.000000] Reading FastQ file /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz;  
[0.003424] Reading FastQ file /localdisk/data/BPSM/Lecture22/6991_2.fastq.gz;  
[6.899199] 3846320 sequences found in total in the paired sequence files  
[6.899213] Done  
[8.502005] Reading read set file paired_k21/Sequences;  
[13.672980] 3846320 sequences found  
[15.371590] Done  
[15.371627] 3846320 sequences in total.  
[15.380404] Writing into roadmap file paired_k21/Roadmaps...  
[17.156269] Inputting sequences...  
[17.156599] Inputting sequence 0 / 3846320  
[23.162178] Inputting sequence 1000000 / 3846320  
[29.552363] Inputting sequence 2000000 / 3846320  
[35.965078] Inputting sequence 3000000 / 3846320  
[41.467466] === Sequences loaded in 24.351325 s  
[41.822045] Done inputting sequences  
[41.822072] Destroying splay table  
[41.847907] Splay table destroyed  
[0.000000] Reading roadmap file paired_k21/Roadmaps  
[7.492867] 3846320 roadmaps read  
[7.494925] Creating insertion markers  
[8.102124] Ordering insertion markers  
[10.273351] Counting preNodes  
[10.716786] 3697105 preNodes counted, creating them now
```

```
[12.432915] Sequence 1000000 / 3846320
[13.280395] Sequence 2000000 / 3846320
[14.090508] Sequence 3000000 / 3846320
[14.765054] Adjusting marker info...
[15.301365] Connecting preNodes
[16.267413] Connecting 1000000 / 3846320
[17.353250] Connecting 2000000 / 3846320
[18.437120] Connecting 3000000 / 3846320
[19.356422] Cleaning up memory
[19.363284] Done creating preGraph
[19.363290] Concatenation...
[19.992649] Renumbering preNodes
[19.992683] Initial preNode count 3697105
[20.029010] Destroyed 3447860 preNodes
[20.029037] Concatenation over!
[20.029038] Clipping short tips off preGraph
[20.078505] Concatenation...
[20.287699] Renumbering preNodes
[20.287733] Initial preNode count 249245
[20.297390] Destroyed 150714 preNodes
[20.297412] Concatenation over!
[20.297413] 79749 tips cut off
[20.297415] 98531 nodes left
[20.299211] Writing into pregraph file paired_k21/PreGraph...
[20.777574] Reading read set file paired_k21/Sequences;
[20.949568] 3846320 sequences found
[22.598807] Done
[24.360274] Reading pre-graph file paired_k21/PreGraph
[24.360378] Graph has 98531 nodes and 3846320 sequences
[24.512498] Scanning pre-graph file paired_k21/PreGraph for k-mers
[24.543472] 6365098 kmers found
[24.741606] Sorting kmer occurrence table ...
[26.207710] Sorting done.
[26.207743] Computing acceleration table...
[26.282786] Computing offsets...
[26.296431] Ghost Threading through reads 0 / 3846320
[30.949262] Ghost Threading through reads 1000000 / 3846320
[35.612258] Ghost Threading through reads 2000000 / 3846320
[40.232664] Ghost Threading through reads 3000000 / 3846320
[44.156010] === Ghost-Threaded in 17.859798 s
[44.156053] Threading through reads 0 / 3846320
[50.822416] Threading through reads 1000000 / 3846320
[57.491168] Threading through reads 2000000 / 3846320
[64.091409] Threading through reads 3000000 / 3846320
[69.668830] === Threaded in 25.512778 s
[69.681094] Correcting graph with cutoff 0.200000
[69.682585] Determining eligible starting points
[69.760774] Done listing starting nodes
[69.760798] Initializing todo lists
[69.776726] Done with initialization
[69.776752] Activating arc lookup table
[69.790536] Done activating arc lookup table
[69.839491] 10000 / 98531 nodes visited
[69.892165] 20000 / 98531 nodes visited
[69.950815] 30000 / 98531 nodes visited
[70.010096] 40000 / 98531 nodes visited
[70.079364] 50000 / 98531 nodes visited
[70.144888] 60000 / 98531 nodes visited
[70.213406] 70000 / 98531 nodes visited
[70.289313] 80000 / 98531 nodes visited
[70.354694] 90000 / 98531 nodes visited
[70.386601] 100000 / 98531 nodes visited
[70.391071] Concatenation...
[70.394549] Renumbering nodes
[70.394563] Initial node count 98531
[70.395275] Removed 66587 null nodes
[70.395284] Concatenation over!
[70.395286] Clipping short tips off graph, drastic
[70.403335] Concatenation...
[70.420136] Renumbering nodes
```

```
[70.420158] Initial node count 31944
[70.420658] Removed 7695 null nodes
[70.420663] Concatenation over!
[70.420664] 24249 nodes left
[70.422310] Writing into graph file paired_k21/Graph...
[71.019750] WARNING: NO COVERAGE CUTOFF PROVIDED
[71.019773] Velvet will probably leave behind many detectable errors
[71.019776] See manual for instructions on how to set the coverage cutoff parameter
[71.019794] Removing contigs with coverage < -1.000000...
[71.020694] Concatenation...
[71.023210] Renumbering nodes
[71.023219] Initial node count 24249
[71.023272] Removed 0 null nodes
[71.023276] Concatenation over!
[71.023732] Concatenation...
[71.025554] Renumbering nodes
[71.025561] Initial node count 24249
[71.025613] Removed 0 null nodes
[71.025617] Concatenation over!
[71.025620] Clipping short tips off graph, drastic
[71.026608] Concatenation...
[71.028468] Renumbering nodes
[71.028478] Initial node count 24249
[71.028539] Removed 0 null nodes
[71.028543] Concatenation over!
[71.028546] 24249 nodes left
[71.028548] WARNING: NO EXPECTED COVERAGE PROVIDED
[71.028556] Velvet will be unable to resolve any repeats
[71.028561] See manual for instructions on how to set the expected coverage parameter
[71.028565] Concatenation...
[71.029587] Renumbering nodes
[71.029595] Initial node count 24249
[71.029619] Removed 0 null nodes
[71.029622] Concatenation over!
[71.029623] Removing reference contigs with coverage < -1.000000...
[71.029939] Concatenation...
[71.030918] Renumbering nodes
[71.030925] Initial node count 24249
[71.030949] Removed 0 null nodes
[71.030951] Concatenation over!
[71.045268] Writing contigs into paired_k21/contigs.fa...
[71.432195] Writing into stats file paired_k21/stats.txt...
[71.501600] Writing into graph file paired_k21/LastGraph...
Final graph has 24249 nodes and n50 of 1524, max 10406, total 5800362, using 0/3846320 rea
[Thurs Nov 27 14:27:42 2025]
Finished job 0.
1 of 1 steps (100%) done
Complete log: .snakemake/log/2024-11-27T142547.849393.snakemake.log
```

Let's add the stats bit to our Snakefile2 file now, call it velvet_snakefile2

```
rule assemble:
    output: "{sample}_paired_k21_contigs.fa"
    input:
        read1 = "/localdisk/data/BPSM/Lecture22/{sample}_1.fastq.gz",
        read2 = "/localdisk/data/BPSM/Lecture22/{sample}_2.fastq.gz"
    shell:
        r"""velveth paired_k21 21 -shortPaired -fastq -separate {input.read1} {input.read2}
        velvetg paired_k21
        mv paired_k21/contigs.fa {output}
        """

rule stats:
    output: "{assem}_contigs_stats.txt"
    input: "{assem}_paired_k21_contigs.fa"
    shell:
        "stats.sh {input} > {output}"
```

Run snakemake again

```
snakemake -pn --cores 1 6991_paired_k21_contigs.fa -s velvet_snakefile2
```

Building DAG of jobs...

Nothing to be done (all requested files are present and up to date).

Why were the stats not being done!?

What is it all about? The **TARGET** !

Look at the command again: what is the target specified?

```
snakemake -pn --cores 1 6991_paired_k21_contigs.fa -s velvet_snakefile2
```

What happens if we give it the correct target (i.e. the output we actually want!)?

```
snakemake -pn --cores 1 6991_paired_k21_contigs_stats.txt -s velvet_snakefile2
```

Building DAG of jobs...

MissingInputException in rule stats in line 12 of \$HOME/Exercises/Lecture22/velvet_snakefile

Missing input files for rule stats:

output: 6991_paired_k21_contigs_stats.txt

wildcards: assem=6991_paired_k21

affected files:

6991_paired_k21_paired_k21_contigs.fa

Clearly, we still haven't quite got things right.

Modify the stats part again

Let's also modify the stats rule to extract two things:

1. the whole set of assembly stats and

2. stats output lines containing the text string "contig"

```
rule assemble:
```

```
output: "{sample}_paired_k21_contigs.fa"
```

```
input:
```

```
read1 = "/localdisk/data/BPSM/Lecture22/{sample}_1.fastq.gz",
```

```
read2 = "/localdisk/data/BPSM/Lecture22/{sample}_2.fastq.gz"
```

```
shell:
```

```
r"""velveth paired_k21 21 -shortPaired -fastq -separate {input.read1} {input.read2}
```

```
velvetg paired_k21
```

```
mv paired_k21/contigs.fa {output}
```

```
"""
```

```
rule stats:
```

```
output:
```

```
full = "{assembly}_contigs_stats.txt",
```

```
grepped = "{assembly}_contigs_stats_grepped.txt"
```

```
input: "{assembly}_contigs.fa"
```

```
shell:
```

```
"""stats.sh {input} > {output.full}
```

```
stats.sh {input} | grep contig > {output.grepped}"""
```

We saved this revised version as **velvet_snakefile3**

Run it for real from the very beginning

delete ALL targets, so they will be made again (i.e. reassembly AND stats)

```
rm -fr paired_k21 ; rm *contigs*
```

```
snakemake -p --cores 1 6991_paired_k21_contigs_stats.txt -s velvet_snakefile3
```

Building DAG of jobs...

Using shell: /usr/bin/bash

Provided cores: 1 (use --cores to define parallelism)

Rules claiming more threads will be scaled down.

Job stats:

job	count	min threads	max threads
assemble	1	1	1
stats	1	1	1
total	2	1	1

Select jobs to execute...

[Thurs Nov 27 14:44:24 2025]

rule assemble:

input: /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz, /localdisk/data/BPSM/Lecture22/

output: 6991_paired_k21_contigs.fa

jobid: 1

reason: Missing output files: 6991_paired_k21_contigs.fa

wildcards: sample=6991

resources: tmpdir=/tmp

```
velveth paired_k21 21 -shortPaired -fastq -separate /localdisk/data/BPSM/Lecture22/6991_1.
```

```
velvetg paired_k21
```

```
mv paired_k21/contigs.fa 6991_paired_k21_contigs.fa
```

```
[0.000000] Reading FastQ file /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz;
```

```
[0.000605] Reading FastQ file /localdisk/data/BPSM/Lecture22/6991_2.fastq.gz;
```

```
[5.832541] 3846320 sequences found in total in the paired sequence files
```

```
[5.832562] Done
```

```
[8.157898] Reading read set file paired_k21/Sequences;
```

```
[12.306337] 3846320 sequences found
```

```
[13.979179] Done
```

```
[13.979202] 3846320 sequences in total.
```

```
[13.980592] Writing into roadmap file paired_k21/Roadmaps...
```

```
[15.756055] Inputting sequences...
```

```
[15.756396] Inputting sequence 0 / 3846320
```

```
[21.767559] Inputting sequence 1000000 / 3846320
```

```
[28.162677] Inputting sequence 2000000 / 3846320
```

```
[34.629885] Inputting sequence 3000000 / 3846320
```

```
[40.164571] === Sequences loaded in 24.449054 s
```

```
[40.887796] Done inputting sequences
```

```
[40.887826] Destroying splay table
```

```
[40.909747] Splay table destroyed
```

```
[0.000000] Reading roadmap file paired_k21/Roadmaps
```

```
[5.113068] 3846320 roadmaps read
```

```
[5.114508] Creating insertion markers
```

```
[5.719964] Ordering insertion markers
```

```
[7.931703] Counting preNodes
```

```
[8.382111] 3697105 preNodes counted, creating them now
```

```
[10.101933] Sequence 1000000 / 3846320
```

```
[10.963676] Sequence 2000000 / 3846320
```

```
[11.776454] Sequence 3000000 / 3846320
```

```
[12.455776] Adjusting marker info...
```

```
[13.074775] Connecting preNodes
```

```
[14.080366] Connecting 1000000 / 3846320
```

```
[15.181999] Connecting 2000000 / 3846320
```

```
[16.284039] Connecting 3000000 / 3846320
```

```
[17.219971] Cleaning up memory
```

```
[17.225716] Done creating preGraph
```

```
[17.225721] Concatenation...
```

```
[17.925405] Renumbering preNodes
```

```
[17.925438] Initial preNode count 3697105
```

```
[17.961073] Destroyed 3447860 preNodes
```

```
[17.961098] Concatenation over!
```

```
[17.961100] Clipping short tips off preGraph
```

```
[18.009944] Concatenation...
```

```
[18.219871] Renumbering preNodes
```

```
[18.219905] Initial preNode count 249245
[18.229503] Destroyed 150714 preNodes
[18.229529] Concatenation over!
[18.229531] 79749 tips cut off
[18.229532] 98531 nodes left
[18.230591] Writing into pregraph file paired_k21/PreGraph...
[18.692881] Reading read set file paired_k21/Sequences;
[18.866552] 3846320 sequences found
[20.515933] Done
[22.288393] Reading pre-graph file paired_k21/PreGraph
[22.290214] Graph has 98531 nodes and 3846320 sequences
[22.486510] Scanning pre-graph file paired_k21/PreGraph for k-mers
[22.517219] 6365098 kmers found
[22.716994] Sorting kmer occurrence table ...
[24.200417] Sorting done.
[24.200446] Computing acceleration table...
[24.275593] Computing offsets...
[24.288968] Ghost Threading through reads 0 / 3846320
[28.949355] Ghost Threading through reads 1000000 / 3846320
[33.607323] Ghost Threading through reads 2000000 / 3846320
[38.225555] Ghost Threading through reads 3000000 / 3846320
[42.149978] === Ghost-Threaded in 17.861231 s
[42.150015] Threading through reads 0 / 3846320
[48.866101] Threading through reads 1000000 / 3846320
[55.578237] Threading through reads 2000000 / 3846320
[62.235929] Threading through reads 3000000 / 3846320
[67.859434] === Threaded in 25.709422 s
[67.871802] Correcting graph with cutoff 0.200000
[67.873317] Determining eligible starting points
[67.954347] Done listing starting nodes
[67.954371] Initializing todo lists
[67.970604] Done with initialization
[67.970624] Activating arc lookup table
[67.984794] Done activating arc lookup table
[68.034196] 10000 / 98531 nodes visited
[68.087245] 20000 / 98531 nodes visited
[68.147110] 30000 / 98531 nodes visited
[68.207403] 40000 / 98531 nodes visited
[68.273391] 50000 / 98531 nodes visited
[68.339871] 60000 / 98531 nodes visited
[68.408849] 70000 / 98531 nodes visited
[68.484941] 80000 / 98531 nodes visited
[68.550674] 90000 / 98531 nodes visited
[68.582633] 100000 / 98531 nodes visited
[68.587018] Concatenation...
[68.590578] Renumbering nodes
[68.590592] Initial node count 98531
[68.591335] Removed 66587 null nodes
[68.591346] Concatenation over!
[68.591348] Clipping short tips off graph, drastic
[68.599715] Concatenation...
[68.616517] Renumbering nodes
[68.616544] Initial node count 31944
[68.617074] Removed 7695 null nodes
[68.617081] Concatenation over!
[68.617082] 24249 nodes left
[68.627017] Writing into graph file paired_k21/Graph...
[69.247273] WARNING: NO COVERAGE CUTOFF PROVIDED
[69.247292] Velvet will probably leave behind many detectable errors
[69.247295] See manual for instructions on how to set the coverage cutoff parameter
[69.247298] Removing contigs with coverage < -1.000000...
[69.248426] Concatenation...
[69.251397] Renumbering nodes
[69.251406] Initial node count 24249
[69.251459] Removed 0 null nodes
[69.251464] Concatenation over!
[69.251943] Concatenation...
[69.254046] Renumbering nodes
[69.254054] Initial node count 24249
[69.254105] Removed 0 null nodes
```

```
[69.254110] Concatenation over!
[69.254113] Clipping short tips off graph, drastic
[69.254672] Concatenation...
[69.255796] Renumbering nodes
[69.255803] Initial node count 24249
[69.255826] Removed 0 null nodes
[69.255828] Concatenation over!
[69.255829] 24249 nodes left
[69.255830] WARNING: NO EXPECTED COVERAGE PROVIDED
[69.255831] Velvet will be unable to resolve any repeats
[69.255832] See manual for instructions on how to set the expected coverage parameter
[69.255834] Concatenation...
[69.256938] Renumbering nodes
[69.256946] Initial node count 24249
[69.256970] Removed 0 null nodes
[69.256971] Concatenation over!
[69.256972] Removing reference contigs with coverage < -1.000000...
[69.257319] Concatenation...
[69.258429] Renumbering nodes
[69.258436] Initial node count 24249
[69.258459] Removed 0 null nodes
[69.258460] Concatenation over!
[69.272180] Writing contigs into paired_k21/contigs.fa...
[69.642736] Writing into stats file paired_k21/stats.txt...
[69.707975] Writing into graph file paired_k21/LastGraph...
Final graph has 24249 nodes and n50 of 1524, max 10406, total 5800362, using 0/3846320 rea
[Thurs Nov 27 14:46:15 2025]
Finished job 1.
1 of 2 steps (50%) done
Select jobs to execute...

[Thurs Nov 27 14:46:15 2025]
rule stats:
  input: 6991_paired_k21_contigs.fa
  output: 6991_paired_k21_contigs_stats.txt, 6991_paired_k21_contigs_stats_grepped.txt
  jobid: 0
  reason: Missing output files: 6991_paired_k21_contigs_stats.txt; Input files updated b
  wildcards: assembly=6991_paired_k21
  resources: tmpdir=/tmp

stats.sh 6991_paired_k21_contigs.fa > 6991_paired_k21_contigs_stats.txt
  stats.sh 6991_paired_k21_contigs.fa | grep contig > 6991_paired_k21_contigs_stats_g
[Thurs Nov 27 14:46:16 2025]
Finished job 0.
2 of 2 steps (100%) done
Complete log: .snakemake/log/2024-11-27T144423.964650.snakemake.log
```

Check some outputs

```
wc -l *k21_contigs*
```

```
127054 6991_paired_k21_contigs.fa
      5 6991_paired_k21_contigs_stats_grepped.txt
     31 6991_paired_k21_contigs_stats.txt
127090 total
```

Cool!

What if we want to try a range of kmer values, and summarise the N50 stats at the end?

What was that about lists, and **expand** ing them to be **wildcards** ?

And can we put our outputs in a folder somewhere separate?



```
# Here's our new velvet_snakefile4: velvetOUT and My_snakeouts are directory names
```



```

samples = ["6991"]
kmers = [19, 20]
rule all_kmer_stats:
    input:
        expand("My_snakeouts/{s}_paired_k{k}_contigs_stats.txt", \
            s=samples, k=kmers )
    output:
        "My_snakeouts/all_stats.txt"
    shell:
        "grep -H 'contig N/L50' {input} > {output}"

rule assemble:
    output: "My_snakeouts/{sample}_paired_k{kmer}_contigs.fa"
    input:
        read1 = "/localdisk/data/BPSM/Lecture22/{sample}_1.fastq.gz",
        read2 = "/localdisk/data/BPSM/Lecture22/{sample}_2.fastq.gz"
    shell:
        r"""velveth velvetOUT {wildcards.kmer} -shortPaired -fastq -separate {input.read1}
        velvetg velvetOUT
        mv velvetOUT/contigs.fa {output}
        """

rule stats:
    output:
        full = "My_snakeouts/{assembly}_contigs_stats.txt",
        grepped = "My_snakeouts/{assembly}_contigs_stats_grepped.txt"
    input: "My_snakeouts/{assembly}_contigs.fa"
    shell:
        """stats.sh {input} > {output.full}
        stats.sh {input} | grep contig > {output.grepped}"""

```

Everything is in `velvet_snakefile4` now, so just (dry) run it...

snakemake -pn -s velvet_snakefile4

Building DAG of jobs...

Job stats:

job	count	min threads	max threads
all_kmer_stats	1	1	1
assemble	2	1	1
stats	2	1	1
total	5	1	1

[Thurs Nov 27 15:05:10 2025]

rule assemble:

```

input: /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz, /localdisk/data/BPSM/Lecture22/
output: My_snakeouts/6991_paired_k19_contigs.fa
jobid: 2
reason: Missing output files: My_snakeouts/6991_paired_k19_contigs.fa
wildcards: sample=6991, kmer=19
resources: tmpdir=/tmp

```

```

velveth velvetOUT 19 -shortPaired -fastq -separate /localdisk/data/BPSM/Lecture22/6991_1.f
    velvetg velvetOUT
    mv velvetOUT/contigs.fa My_snakeouts/6991_paired_k19_contigs.fa

```

[Thurs Nov 27 15:05:10 2025]

rule assemble:

```

input: /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz, /localdisk/data/BPSM/Lecture22/
output: My_snakeouts/6991_paired_k20_contigs.fa
jobid: 4
reason: Missing output files: My_snakeouts/6991_paired_k20_contigs.fa
wildcards: sample=6991, kmer=20
resources: tmpdir=/tmp

```

```

velveth velvetOUT 20 -shortPaired -fastq -separate /localdisk/data/BPSM/Lecture22/6991_1.f

```



```
velvetg velvetOUT
mv velvetOUT/contigs.fa My_snakeouts/6991_paired_k20_contigs.fa
```

[Thurs Nov 27 15:05:10 2025]

rule stats:

```
input: My_snakeouts/6991_paired_k19_contigs.fa
output: My_snakeouts/6991_paired_k19_contigs_stats.txt, My_snakeouts/6991_paired_k19_c
jobid: 1
reason: Missing output files: My_snakeouts/6991_paired_k19_contigs_stats.txt; Input fi
wildcards: assembly=6991_paired_k19
resources: tmpdir=/tmp
```

```
stats.sh My_snakeouts/6991_paired_k19_contigs.fa > My_snakeouts/6991_paired_k19_contigs_st
stats.sh My_snakeouts/6991_paired_k19_contigs.fa | grep contig > My_snakeouts/6991_
```

[Thurs Nov 27 15:05:10 2025]

rule stats:

```
input: My_snakeouts/6991_paired_k20_contigs.fa
output: My_snakeouts/6991_paired_k20_contigs_stats.txt, My_snakeouts/6991_paired_k20_c
jobid: 3
reason: Missing output files: My_snakeouts/6991_paired_k20_contigs_stats.txt; Input fi
wildcards: assembly=6991_paired_k20
resources: tmpdir=/tmp
```

```
stats.sh My_snakeouts/6991_paired_k20_contigs.fa > My_snakeouts/6991_paired_k20_contigs_st
stats.sh My_snakeouts/6991_paired_k20_contigs.fa | grep contig > My_snakeouts/6991_
```

[Thurs Nov 27 15:05:10 2025]

rule all_kmer_stats:

```
input: My_snakeouts/6991_paired_k19_contigs_stats.txt, My_snakeouts/6991_paired_k20_co
output: My_snakeouts/all_stats.txt
jobid: 0
reason: Missing output files: My_snakeouts/all_stats.txt; Input files updated by anoth
resources: tmpdir=/tmp
```

```
grep -H 'contig N/L50' My_snakeouts/6991_paired_k19_contigs_stats.txt My_snakeouts/6991_pa
Job stats:
```

job	count	min threads	max threads
all_kmer_stats	1	1	1
assemble	2	1	1
stats	2	1	1
total	5	1	1

Reasons:

```
(check individual jobs above for details)
input files updated by another job:
  all_kmer_stats, stats
output files have to be generated:
  all_kmer_stats, assemble, stats
```

This was a dry-run (flag -n). The order of jobs does not reflect the order of execution.



Seems OK, lets run it, and add some more cores for snakemake!

```
snakemake -p --cores 12 -s velvet_snakefile4
```

```
[0.000000] Reading FastQ file /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz;
[0.000614] Reading FastQ file /localdisk/data/BPSM/Lecture22/6991_2.fastq.gz;
[0.000000] Velvet can't work with even length k-mers, such as 20. We'll use 19 instead, if
[0.011604] Reading FastQ file /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz;
[0.012227] Reading FastQ file /localdisk/data/BPSM/Lecture22/6991_2.fastq.gz;
...
```

Sometimes things dont work...

```
Building DAG of jobs...
Using shell: /usr/bin/bash
Provided cores: 12
Rules claiming more threads will be scaled down.
```

Job stats:

job	count	min threads	max threads
all_kmer_stats	1	1	1
assemble	2	1	1
stats	2	1	1
total	5	1	1

Select jobs to execute...

[Thurs Nov 27 15:06:47 2025]

rule assemble:

```
input: /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz, /localdisk/data/BPSM/Lecture22/
output: My_snakeouts/6991_paired_k19_contigs.fa
jobid: 2
reason: Missing output files: My_snakeouts/6991_paired_k19_contigs.fa
wildcards: sample=6991, kmer=19
resources: tmpdir=/tmp
```

```
velveth velvetOUT 19 -shortPaired -fastq -separate /localdisk/data/BPSM/Lecture22/6991_1.f
velvetg velvetOUT
mv velvetOUT/contigs.fa My_snakeouts/6991_paired_k19_contigs.fa
```

[Thurs Nov 27 15:06:47 2025]

rule assemble:

```
input: /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz, /localdisk/data/BPSM/Lecture22/
output: My_snakeouts/6991_paired_k20_contigs.fa
jobid: 4
reason: Missing output files: My_snakeouts/6991_paired_k20_contigs.fa
wildcards: sample=6991, kmer=20
resources: tmpdir=/tmp
```

```
velveth velvetOUT 20 -shortPaired -fastq -separate /localdisk/data/BPSM/Lecture22/6991_1.f
velvetg velvetOUT
mv velvetOUT/contigs.fa My_snakeouts/6991_paired_k20_contigs.fa
```

```
[0.000000] Reading FastQ file /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz;
[0.000614] Reading FastQ file /localdisk/data/BPSM/Lecture22/6991_2.fastq.gz;
[0.000000] Velvet can't work with even length k-mers, such as 20. We'll use 19 instead, if
[0.011604] Reading FastQ file /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz;
[0.012227] Reading FastQ file /localdisk/data/BPSM/Lecture22/6991_2.fastq.gz;
[11.241989] 3846320 sequences found in total in the paired sequence files
[11.242002] Done
[12.038699] 3846320 sequences found in total in the paired sequence files
[12.038715] Done
[12.089673] Reading read set file velvetOUT/Sequences;
[12.097238] Reading read set file velvetOUT/Sequences;
...
[75.604153] Concatenation over!
[75.619493] Writing contigs into velvetOUT/contigs.fa...
[76.020532] Writing into stats file velvetOUT/stats.txt...
[76.220131] Writing into graph file velvetOUT/LastGraph...
Final graph has 35909 nodes and n50 of 832, max 6029, total 5871961, using 0/3846320 reads
[Thurs Nov 27 15:09:02 2025]
Finished job 4.
1 of 5 steps (20%) done
Select jobs to execute...
```

[Thurs Nov 27 15:09:02 2025]

rule stats:

```
input: My_snakeouts/6991_paired_k20_contigs.fa
output: My_snakeouts/6991_paired_k20_contigs_stats.txt, My_snakeouts/6991_paired_k20_c
jobid: 3
reason: Missing output files: My_snakeouts/6991_paired_k20_contigs_stats.txt; Input fi
wildcards: assembly=6991_paired_k20
resources: tmpdir=/tmp
```

```
stats.sh My_snakeouts/6991_paired_k20_contigs.fa > My_snakeouts/6991_paired_k20_contigs_st
stats.sh My_snakeouts/6991_paired_k20_contigs.fa | grep contig > My_snakeouts/6991_
```

Final graph has 35909 nodes and n50 of 832, max 6029, total 5871961, using 0/3846320 reads
mv: cannot stat 'velvetOUT/contigs.fa': No such file or directory

[Thurs Nov 27 15:09:02 2025]

Error in rule assemble:

```
jobid: 2
input: /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz, /localdisk/data/BPSM/Lecture22/
output: My_snakeouts/6991_paired_k19_contigs.fa
shell:
    velveth velvetOUT 19 -shortPaired -fastq -separate /localdisk/data/BPSM/Lecture22/
    velvetg velvetOUT
    mv velvetOUT/contigs.fa My_snakeouts/6991_paired_k19_contigs.fa
```

(one of the commands exited with non-zero exit code; note that snakemake uses bash

[Thurs Nov 27 15:09:02 2025]

Finished job 3.

2 of 5 steps (40%) done

Shutting down, this might take some time.

Exiting because a job execution failed. Look above for error message

Complete log: .snakemake/log/2024-11-27T150646.950304.snakemake.log

ls -alR velvetOUT

```
velvetOUT:
total 598648
drwxr-xr-x 1 aivens2 g_aivens2      260 Nov 27 15:09 .
drwxr-xr-x 1 aivens2 g_aivens2      924 Nov 27 15:06 ..
-rw-r--r-- 1 aivens2 g_aivens2 13535562 Nov 27 15:09 Graph
-rw-r--r-- 1 aivens2 g_aivens2 13535562 Nov 27 15:09 LastGraph
-rw-r--r-- 1 aivens2 g_aivens2      1756 Nov 27 15:09 Log
-rw-r--r-- 1 aivens2 g_aivens2 12325785 Nov 27 15:08 PreGraph
-rw-r--r-- 1 aivens2 g_aivens2 206193771 Nov 27 15:07 Roadmaps
-rw-r--r-- 1 aivens2 g_aivens2 365067914 Nov 27 15:06 Sequences
-rw-r--r-- 1 aivens2 g_aivens2   2332147 Nov 27 15:09 stats.txt
```

ls -alR My_snakeouts

```
My_snakeouts:
total 7020
drwxr-xr-x 1 aivens2 g_aivens2      212 Nov 27 15:09 .
drwxr-xr-x 1 aivens2 g_aivens2      924 Nov 27 15:06 ..
-rw-r--r-- 1 aivens2 g_aivens2 7170263 Nov 27 15:09 6991_paired_k20_contigs.fa
-rw-r--r-- 1 aivens2 g_aivens2      240 Nov 27 15:09 6991_paired_k20_contigs_stats_grepped.
-rw-r--r-- 1 aivens2 g_aivens2   1586 Nov 27 15:09 6991_paired_k20_contigs_stats.txt
```



That usually gives a segmentation fault as the error!

Can we finish it off? `snakemake` looks for missing targets...

snakemake -p --cores 1 -s velvet_snakefile4

Building DAG of jobs...

Using shell: /usr/bin/bash

Provided cores: 1 (use --cores to define parallelism)

Rules claiming more threads will be scaled down.

Job stats:

job	count	min threads	max threads
all_kmer_stats	1	1	1
assemble	1	1	1
stats	1	1	1
total	3	1	1

Select jobs to execute...

[Thurs Nov 27 16:16:59 2025]

rule assemble:

```
input: /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz, /localdisk/data/BPSM/Lecture22/
output: My_snakeouts/6991_paired_k19_contigs.fa
jobid: 2
reason: Missing output files: My_snakeouts/6991_paired_k19_contigs.fa
```

```
wildcards: sample=6991, kmer=19
resources: tmpdir=/tmp

velveth velvetOUT 19 -shortPaired -fastq -separate /localdisk/data/BPSM/Lecture22/6991_1.f
  velvetg velvetOUT
  mv velvetOUT/contigs.fa My_snakeouts/6991_paired_k19_contigs.fa

[0.000000] Reading FastQ file /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz;
[0.000559] Reading FastQ file /localdisk/data/BPSM/Lecture22/6991_2.fastq.gz;
[5.783477] 3846320 sequences found in total in the paired sequence files
[5.783492] Done
[9.676770] Reading read set file velvetOUT/Sequences;
[9.840045] 3846320 sequences found
[11.490109] Done
[11.490137] 3846320 sequences in total.
[11.513688] Writing into roadmap file velvetOUT/Roadmaps...
[13.316464] Inputting sequences...
[13.316819] Inputting sequence 0 / 3846320
...
[77.557681] Writing into graph file velvetOUT/Graph...
[78.201503] WARNING: NO COVERAGE CUTOFF PROVIDED
[78.201524] Velvet will probably leave behind many detectable errors
[78.201526] See manual for instructions on how to set the coverage cutoff parameter
[78.201529] Removing contigs with coverage < -1.000000...
[78.203269] Concatenation...
...
[78.238803] Writing contigs into velvetOUT/contigs.fa...
[78.623931] Writing into stats file velvetOUT/stats.txt...
[78.719967] Writing into graph file velvetOUT/LastGraph...
Final graph has 35909 nodes and n50 of 832, max 6029, total 5871961, using 0/3846320 reads
[Thurs Nov 27 16:18:59 2025]
Finished job 2.
1 of 3 steps (33%) done
Select jobs to execute...

[Thurs Nov 27 16:18:59 2025]
rule stats:
  input: My_snakeouts/6991_paired_k19_contigs.fa
  output: My_snakeouts/6991_paired_k19_contigs_stats.txt, My_snakeouts/6991_paired_k19_c
  jobid: 1
  reason: Missing output files: My_snakeouts/6991_paired_k19_contigs_stats.txt; Input fi
  wildcards: assembly=6991_paired_k19
  resources: tmpdir=/tmp

stats.sh My_snakeouts/6991_paired_k19_contigs.fa > My_snakeouts/6991_paired_k19_contigs_st
  stats.sh My_snakeouts/6991_paired_k19_contigs.fa | grep contig > My_snakeouts/6991_
[Thurs Nov 27 16:18:59 2025]
Finished job 1.
2 of 3 steps (67%) done
Select jobs to execute...

[Thurs Nov 27 16:18:59 2025]
rule all_kmer_stats:
  input: My_snakeouts/6991_paired_k19_contigs_stats.txt, My_snakeouts/6991_paired_k20_co
  output: My_snakeouts/all_stats.txt
  jobid: 0
  reason: Missing output files: My_snakeouts/all_stats.txt; Input files updated by anoth
  resources: tmpdir=/tmp

grep -H 'contig N/L50' My_snakeouts/6991_paired_k19_contigs_stats.txt My_snakeouts/6991_pa
[Thurs Nov 27 16:19:00 2025]
Finished job 0.
3 of 3 steps (100%) done
Complete log: .snakemake/log/2024-11-27T161659.231693.snakemake.log
```

Looking hopeful...!

Is -aIR My_snakeouts

My_snakeouts:

total 14036

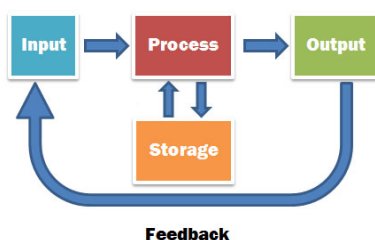
```
drwxr-xr-x 1 aivens2 g_aivens2      400 Nov 27 16:18 .
drwxr-xr-x 1 aivens2 g_aivens2      924 Nov 27 15:06 ..
-rw-r--r-- 1 aivens2 g_aivens2 7170263 Nov 27 16:18 6991_paired_k19_contigs.fa
-rw-r--r-- 1 aivens2 g_aivens2      240 Nov 27 16:18 6991_paired_k19_contigs_stats_grepped.
-rw-r--r-- 1 aivens2 g_aivens2     1586 Nov 27 16:18 6991_paired_k19_contigs_stats.txt
-rw-r--r-- 1 aivens2 g_aivens2 7170263 Nov 27 15:09 6991_paired_k20_contigs.fa
-rw-r--r-- 1 aivens2 g_aivens2      240 Nov 27 15:09 6991_paired_k20_contigs_stats_grepped.
-rw-r--r-- 1 aivens2 g_aivens2     1586 Nov 27 15:09 6991_paired_k20_contigs_stats.txt
-rw-r--r-- 1 aivens2 g_aivens2      186 Nov 27 16:19 all_stats.txt
```

🏠 Exercises: your turn!

Make a directory within your Exercises space for today's work, and change directory to it.

```
mkdir ~/Exercises/Lecture22
```

```
cd ~/Exercises/Lecture22
```



Using what we have done today as a guide:

1. 🏠 use conda to install snakemake in an environment, add matplotlib too...
2. install BBMap tools : done for you already ✅
3. git clone velvet to install it : done for you already ✅
4. 🏠 ensure that you can get snakemake working for Snakefile1, etc.
5. 🏠 maybe have a go at getting a snakemake for velvet assemblies working for a range of kmer sizes (I suggest 19 to 31 in steps of 2). What effect does that have on contigs sizes?
6. 🏠 maybe have a go at getting a snakemake for velvet assemblies working for the *Trypanosoma congolense* transcriptome data: a very different prospect (why?)

Files are located here: `/localdisk/data/BPSM/Lecture22/` (including a set of reads for the *Trypanosoma congolense* transcriptome data)

Interested in how to use Nextflow, an alternative to Snakemake? Check out these links:

1. [Excellent step by step course](#)
2. [The Nextflow web site](#)

A final re-thanks to Edinburgh Genomics: Tim Booth (Hywel Dunn-Davies, Katie Emelionova and Mark Blaxter) for assistance and bacterial genome sequence data used for the velvet assemblies!

Give feedback
Ask AI for help



A longer version of this Snakemake session is available as part of the [EdDaSH Edinburgh Data Carpentries](#) workshop series.

Exercise possible answers and git stuff below as usual....!

POSSIBLE ANSWERS

1. ensure that you can get snakemake working for Snakefile1, Snakefile2 etc
2. have a go at getting velvet to work for a range of kmer sizes (I suggest 19 to 31 in steps of 2). What effect does that have?

Here's the snakefile called SnakefileEx1 (it was provided for you!)

```
samples = ["6991"]
kmers = range(19, 32, 2)

rule all_kmer_stats:
    input:
        expand("ExerciseSnakeout/{s}_paired_k{k}_contigs_stats.txt", \
              s=samples, k=kmers )
    output:
        "ExerciseSnakeout/all_stats.txt"
    shell:
        "grep -H 'contig N/L50' {input} > {output}"

rule assemble:
    output: "ExerciseSnakeout/{sample}_paired_k{kmer}_contigs.fa"
    input:
        read1 = "/localdisk/data/BPSM/Lecture22/{sample}_1.fastq.gz",
        read2 = "/localdisk/data/BPSM/Lecture22/{sample}_2.fastq.gz"
    shell:
        r"""velveth ExercisevelvetOUT {wildcards.kmer} -shortPaired -fastq -separate {i}
        velvetg ExercisevelvetOUT
        mv ExercisevelvetOUT/contigs.fa {output}
        """

rule stats:
    output:
        full      = "ExerciseSnakeout/{assembly}_contigs_stats.txt",
        grepped   = "ExerciseSnakeout/{assembly}_contigs_stats_grepped.txt"
    input: "ExerciseSnakeout/{assembly}_contigs.fa"
```



```
shell:
"""stats.sh {input} > {output.full}
stats.sh {input} | grep contig > {output.grepped}"""
```

Try it out as a DRY RUN, the snakefile is called SnakefileEx1

snakemake -pn --cores 1 -s SnakefileEx1

Building DAG of jobs...

Job stats:

job	count	min threads	max threads
all_kmer_stats	1	1	1
assemble	7	1	1
stats	7	1	1
total	15	1	1

[Thurs Nov 27 16:31:27 2025]

rule assemble:

```
input: /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz, /localdisk/data/BPSM/Lectu
output: ExerciseSnakeout/6991_paired_k25_contigs.fa
jobid: 8
reason: Missing output files: ExerciseSnakeout/6991_paired_k25_contigs.fa
wildcards: sample=6991, kmer=25
resources: tmpdir=/tmp
```

```
velveth ExercisevelvetOUT 25 -shortPaired -fastq -separate /localdisk/data/BPSM/Lectu
velvetg ExercisevelvetOUT
mv ExercisevelvetOUT/contigs.fa ExerciseSnakeout/6991_paired_k25_contigs.fa
```

[Thurs Nov 27 16:31:27 2025]

rule assemble:

```
input: /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz, /localdisk/data/BPSM/Lectu
output: ExerciseSnakeout/6991_paired_k27_contigs.fa
jobid: 10
reason: Missing output files: ExerciseSnakeout/6991_paired_k27_contigs.fa
wildcards: sample=6991, kmer=27
resources: tmpdir=/tmp
```

```
velveth ExercisevelvetOUT 27 -shortPaired -fastq -separate /localdisk/data/BPSM/Lectu
velvetg ExercisevelvetOUT
mv ExercisevelvetOUT/contigs.fa ExerciseSnakeout/6991_paired_k27_contigs.fa
```

[Thurs Nov 27 16:31:27 2025]

rule assemble:

```
input: /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz, /localdisk/data/BPSM/Lectu
output: ExerciseSnakeout/6991_paired_k29_contigs.fa
jobid: 12
reason: Missing output files: ExerciseSnakeout/6991_paired_k29_contigs.fa
wildcards: sample=6991, kmer=29
resources: tmpdir=/tmp
```

```
velveth ExercisevelvetOUT 29 -shortPaired -fastq -separate /localdisk/data/BPSM/Lectu
velvetg ExercisevelvetOUT
mv ExercisevelvetOUT/contigs.fa ExerciseSnakeout/6991_paired_k29_contigs.fa
```

[Thurs Nov 27 16:31:27 2025]

rule assemble:

```
input: /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz, /localdisk/data/BPSM/Lectu
output: ExerciseSnakeout/6991_paired_k31_contigs.fa
jobid: 14
reason: Missing output files: ExerciseSnakeout/6991_paired_k31_contigs.fa
wildcards: sample=6991, kmer=31
resources: tmpdir=/tmp
```

```
velveth ExercisevelvetOUT 31 -shortPaired -fastq -separate /localdisk/data/BPSM/Lectu
```

```
velvetg ExercisevelvetOUT
mv ExercisevelvetOUT/contigs.fa ExerciseSnakeout/6991_paired_k31_contigs.fa
```

[Thurs Nov 27 16:31:27 2025]

```
rule assemble:
  input: /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz, /localdisk/data/BPSM/Lecture22/6991_2.fastq.gz
  output: ExerciseSnakeout/6991_paired_k19_contigs.fa
  jobid: 2
  reason: Missing output files: ExerciseSnakeout/6991_paired_k19_contigs.fa
  wildcards: sample=6991, kmer=19
  resources: tmpdir=/tmp
```

```
velveth ExercisevelvetOUT 19 -shortPaired -fastq -separate /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz /localdisk/data/BPSM/Lecture22/6991_2.fastq.gz
velvetg ExercisevelvetOUT
mv ExercisevelvetOUT/contigs.fa ExerciseSnakeout/6991_paired_k19_contigs.fa
```

[Thurs Nov 27 16:31:27 2025]

```
rule assemble:
  input: /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz, /localdisk/data/BPSM/Lecture22/6991_2.fastq.gz
  output: ExerciseSnakeout/6991_paired_k21_contigs.fa
  jobid: 4
  reason: Missing output files: ExerciseSnakeout/6991_paired_k21_contigs.fa
  wildcards: sample=6991, kmer=21
  resources: tmpdir=/tmp
```

```
velveth ExercisevelvetOUT 21 -shortPaired -fastq -separate /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz /localdisk/data/BPSM/Lecture22/6991_2.fastq.gz
velvetg ExercisevelvetOUT
mv ExercisevelvetOUT/contigs.fa ExerciseSnakeout/6991_paired_k21_contigs.fa
```

[Thurs Nov 27 16:31:27 2025]

```
rule assemble:
  input: /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz, /localdisk/data/BPSM/Lecture22/6991_2.fastq.gz
  output: ExerciseSnakeout/6991_paired_k23_contigs.fa
  jobid: 6
  reason: Missing output files: ExerciseSnakeout/6991_paired_k23_contigs.fa
  wildcards: sample=6991, kmer=23
  resources: tmpdir=/tmp
```

```
velveth ExercisevelvetOUT 23 -shortPaired -fastq -separate /localdisk/data/BPSM/Lecture22/6991_1.fastq.gz /localdisk/data/BPSM/Lecture22/6991_2.fastq.gz
velvetg ExercisevelvetOUT
mv ExercisevelvetOUT/contigs.fa ExerciseSnakeout/6991_paired_k23_contigs.fa
```

[Thurs Nov 27 16:31:27 2025]

```
rule stats:
  input: ExerciseSnakeout/6991_paired_k27_contigs.fa
  output: ExerciseSnakeout/6991_paired_k27_contigs_stats.txt, ExerciseSnakeout/6991_paired_k27_contigs_stats.html
  jobid: 9
  reason: Missing output files: ExerciseSnakeout/6991_paired_k27_contigs_stats.txt; ExerciseSnakeout/6991_paired_k27_contigs_stats.html
  wildcards: assembly=6991_paired_k27
  resources: tmpdir=/tmp
```

```
stats.sh ExerciseSnakeout/6991_paired_k27_contigs.fa > ExerciseSnakeout/6991_paired_k27_contigs_stats.txt
stats.sh ExerciseSnakeout/6991_paired_k27_contigs.fa | grep contig > ExerciseSnakeout/6991_paired_k27_contigs_stats.html
```

[Thurs Nov 27 16:31:27 2025]

```
rule stats:
  input: ExerciseSnakeout/6991_paired_k29_contigs.fa
  output: ExerciseSnakeout/6991_paired_k29_contigs_stats.txt, ExerciseSnakeout/6991_paired_k29_contigs_stats.html
  jobid: 11
  reason: Missing output files: ExerciseSnakeout/6991_paired_k29_contigs_stats.txt; ExerciseSnakeout/6991_paired_k29_contigs_stats.html
  wildcards: assembly=6991_paired_k29
  resources: tmpdir=/tmp
```

```
stats.sh ExerciseSnakeout/6991_paired_k29_contigs.fa > ExerciseSnakeout/6991_paired_k29_contigs_stats.txt
stats.sh ExerciseSnakeout/6991_paired_k29_contigs.fa | grep contig > ExerciseSnakeout/6991_paired_k29_contigs_stats.html
```

```
[Thurs Nov 27 16:31:27 2025]
rule stats:
  input: ExerciseSnakeout/6991_paired_k31_contigs.fa
  output: ExerciseSnakeout/6991_paired_k31_contigs_stats.txt, ExerciseSnakeout/6991
  jobid: 13
  reason: Missing output files: ExerciseSnakeout/6991_paired_k31_contigs_stats.txt;
  wildcards: assembly=6991_paired_k31
  resources: tmpdir=/tmp

stats.sh ExerciseSnakeout/6991_paired_k31_contigs.fa > ExerciseSnakeout/6991_paired_k
stats.sh ExerciseSnakeout/6991_paired_k31_contigs.fa | grep contig > ExerciseS

[Thurs Nov 27 16:31:27 2025]
rule stats:
  input: ExerciseSnakeout/6991_paired_k19_contigs.fa
  output: ExerciseSnakeout/6991_paired_k19_contigs_stats.txt, ExerciseSnakeout/6991
  jobid: 1
  reason: Missing output files: ExerciseSnakeout/6991_paired_k19_contigs_stats.txt;
  wildcards: assembly=6991_paired_k19
  resources: tmpdir=/tmp

stats.sh ExerciseSnakeout/6991_paired_k19_contigs.fa > ExerciseSnakeout/6991_paired_k
stats.sh ExerciseSnakeout/6991_paired_k19_contigs.fa | grep contig > ExerciseS

[Thurs Nov 27 16:31:27 2025]
rule stats:
  input: ExerciseSnakeout/6991_paired_k21_contigs.fa
  output: ExerciseSnakeout/6991_paired_k21_contigs_stats.txt, ExerciseSnakeout/6991
  jobid: 3
  reason: Missing output files: ExerciseSnakeout/6991_paired_k21_contigs_stats.txt;
  wildcards: assembly=6991_paired_k21
  resources: tmpdir=/tmp

stats.sh ExerciseSnakeout/6991_paired_k21_contigs.fa > ExerciseSnakeout/6991_paired_k
stats.sh ExerciseSnakeout/6991_paired_k21_contigs.fa | grep contig > ExerciseS

[Thurs Nov 27 16:31:27 2025]
rule stats:
  input: ExerciseSnakeout/6991_paired_k23_contigs.fa
  output: ExerciseSnakeout/6991_paired_k23_contigs_stats.txt, ExerciseSnakeout/6991
  jobid: 5
  reason: Missing output files: ExerciseSnakeout/6991_paired_k23_contigs_stats.txt;
  wildcards: assembly=6991_paired_k23
  resources: tmpdir=/tmp

stats.sh ExerciseSnakeout/6991_paired_k23_contigs.fa > ExerciseSnakeout/6991_paired_k
stats.sh ExerciseSnakeout/6991_paired_k23_contigs.fa | grep contig > ExerciseS

[Thurs Nov 27 16:31:27 2025]
rule stats:
  input: ExerciseSnakeout/6991_paired_k25_contigs.fa
  output: ExerciseSnakeout/6991_paired_k25_contigs_stats.txt, ExerciseSnakeout/6991
  jobid: 7
  reason: Missing output files: ExerciseSnakeout/6991_paired_k25_contigs_stats.txt;
  wildcards: assembly=6991_paired_k25
  resources: tmpdir=/tmp

stats.sh ExerciseSnakeout/6991_paired_k25_contigs.fa > ExerciseSnakeout/6991_paired_k
stats.sh ExerciseSnakeout/6991_paired_k25_contigs.fa | grep contig > ExerciseS

[Thurs Nov 27 16:31:27 2025]
rule all_kmer_stats:
  input: ExerciseSnakeout/6991_paired_k19_contigs_stats.txt, ExerciseSnakeout/6991
  output: ExerciseSnakeout/all_stats.txt
  jobid: 0
  reason: Missing output files: ExerciseSnakeout/all_stats.txt; Input files updated
  resources: tmpdir=/tmp

grep -H 'contig N/L50' ExerciseSnakeout/6991_paired_k19_contigs_stats.txt ExerciseS
Job stats:
```

job	count	min threads	max threads
all_kmer_stats	1	1	1
assemble	7	1	1
stats	7	1	1
total	15	1	1

Reasons:

- (check individual jobs above for details)
- input files updated by another job:
 - all_kmer_stats, stats
- output files have to be generated:
 - all_kmer_stats, assemble, stats

This was a dry-run (flag -n). The order of jobs does not reflect the order of execution.

The big run for real!

```
snakemake -p --cores 1 -s SnakefileEx1
```

... lots ...

[Thurs Nov 27 16:44:13 2025]
Finished job 0.
15 of 15 steps (100%) done
Complete log: .snakemake/log/2024-11-27T163232.543073.snakemake.log

About 15 minutes later....

ls -altrR ExerciseSnakeout

ExerciseSnakeout/:
total 44388

drwxr-xr-x	1	aivens2	g_aivens2	996	Nov 27 16:32	..
-rw-r--r--	1	aivens2	g_aivens2	6275975	Nov 27 16:34	6991_paired_k27_contigs.fa
-rw-r--r--	1	aivens2	g_aivens2	6129170	Nov 27 16:35	6991_paired_k29_contigs.fa
-rw-r--r--	1	aivens2	g_aivens2	6006388	Nov 27 16:36	6991_paired_k31_contigs.fa
-rw-r--r--	1	aivens2	g_aivens2	6796701	Nov 27 16:38	6991_paired_k21_contigs.fa
-rw-r--r--	1	aivens2	g_aivens2	6575103	Nov 27 16:40	6991_paired_k23_contigs.fa
-rw-r--r--	1	aivens2	g_aivens2	6410482	Nov 27 16:42	6991_paired_k25_contigs.fa
-rw-r--r--	1	aivens2	g_aivens2	7170263	Nov 27 16:44	6991_paired_k19_contigs.fa
-rw-r--r--	1	aivens2	g_aivens2	1586	Nov 27 16:44	6991_paired_k19_contigs_stats.txt
-rw-r--r--	1	aivens2	g_aivens2	240	Nov 27 16:44	6991_paired_k19_contigs_stats_gre
-rw-r--r--	1	aivens2	g_aivens2	1674	Nov 27 16:44	6991_paired_k21_contigs_stats.txt
-rw-r--r--	1	aivens2	g_aivens2	245	Nov 27 16:44	6991_paired_k21_contigs_stats_gre
-rw-r--r--	1	aivens2	g_aivens2	1674	Nov 27 16:44	6991_paired_k23_contigs_stats.txt
-rw-r--r--	1	aivens2	g_aivens2	245	Nov 27 16:44	6991_paired_k23_contigs_stats_gre
-rw-r--r--	1	aivens2	g_aivens2	1674	Nov 27 16:44	6991_paired_k25_contigs_stats.txt
-rw-r--r--	1	aivens2	g_aivens2	245	Nov 27 16:44	6991_paired_k25_contigs_stats_gre
-rw-r--r--	1	aivens2	g_aivens2	1670	Nov 27 16:44	6991_paired_k27_contigs_stats.txt
-rw-r--r--	1	aivens2	g_aivens2	243	Nov 27 16:44	6991_paired_k27_contigs_stats_gre
-rw-r--r--	1	aivens2	g_aivens2	1674	Nov 27 16:44	6991_paired_k29_contigs_stats.txt
-rw-r--r--	1	aivens2	g_aivens2	245	Nov 27 16:44	6991_paired_k29_contigs_stats_gre
-rw-r--r--	1	aivens2	g_aivens2	1676	Nov 27 16:44	6991_paired_k31_contigs_stats.txt
-rw-r--r--	1	aivens2	g_aivens2	246	Nov 27 16:44	6991_paired_k31_contigs_stats_gre
drwxr-xr-x	1	aivens2	g_aivens2	1180	Nov 27 16:44	.
-rw-r--r--	1	aivens2	g_aivens2	704	Nov 27 16:44	all_stats.txt

 git and GitHub time!

Local repository actions (the absolute minimum set of commands required here...)

```
git st
```

On branch master

Changes not staged for commit:

```
(use "git add/rm <file>..." to update what will be committed)
(use "git restore <file>..." to discard changes in working directory)
deleted:    ../Lecture04/CodeFiles.tar.gz
modified:   ../Lecture06/run_this_script_v1
```

Untracked files:

```
(use "git add <file>..." to include in what will be committed)
../BN.out
../BN.sh
../Lecture04/Als_ICA/
../Lecture04/all_the_things_I_did
../Lecture04/outs/
../Lecture05/Country.Mozambique.details
../Lecture07/Cosmoscarta.nuc.fa
../Lecture07/Cosmoscarta.nuc.gis
../Lecture07/myfileofaccessions.txt
../Lecture12/AJ223353.fasta
../Lecture12/AJ223353.genbank
../Lecture12/AJ223353_noheader.fasta
../Lecture12/AJ223353_noheader.fasta2
../Lecture12/AJ223353_noheader.fasta3
../Lecture12/All_exons.fasta
../Lecture12/All_noncodings.fasta
../Lecture12/local_exon01.fasta
../Lecture12/local_exon02.fasta
../Lecture12/local_intron01.fasta
../Lecture12/my_dna.txt
../Lecture12/plain_genomic_seq.txt
../Lecture12/quick_blast_check.out
../Lecture12/remote_exon01.fasta
../Lecture12/remote_noncoding01.fasta
../Lecture12/remote_noncoding02.fasta
../Lecture12/shutilversion.txt
../Lecture13/100_199/
../Lecture13/200_299/
../Lecture13/300_399/
../Lecture13/400_499/
../Lecture13/500_599/
../Lecture13/600_699/
../Lecture13/700_799/
../Lecture13/800_899/
../Lecture13/900_999/
../Lecture13/Cleaned_sequences.txt
../Lecture13/Cleaned_sequences2.txt
../Lecture13/Cleaned_sequences3.txt
../Lecture13/Exercise2_coding_sequence.fasta
../Lecture13/Lecture13.tar.gz
../Lecture13/dna_files/
../Lecture13/exons.txt
../Lecture13/genomic_dna2.txt
../Lecture13/input.txt
../Lecture13/remote_exon01.fasta
../Lecture14/__pycache__/
../Lecture14/data.csv
../Lecture16/__pycache__/
../Lecture17/All_the_Arabidopsis_ones.tsv
../Lecture17/eukaryotes.txt
../Lecture17/playtime/
../Lecture19/Chart_01.png
../Lecture19/Chart_02.png
../Lecture19/Chart_03.png
../Lecture19/Chart_04.png
../Lecture19/Chart_05.png
../Lecture19/Chart_06.png
../Lecture19/Chart_07.png
../Lecture19/Chart_08.png
../Lecture19/Chart_09.png
```

```
../Lecture19/Chart_10.png
../Lecture19/Chart_11.png
../Lecture19/Chart_12.png
../Lecture19/Chart_13.png
../Lecture19/Chart_14.png
../Lecture19/Chart_15.png
../Lecture19/Chart_16.png
../Lecture19/Chart_16A.png
../Lecture19/Chart_77.png
../Lecture20/Arthropoda_ATP6_complete_outputs.txt
../Lecture20/Bos_taurus_cytochrome_c_oxidase_subunit_I_(mitochondrion)_complete_ou
../Lecture20/Mammalia_COX1_complete_outputs.txt
../Lecture20/Nematoda_COX1_complete_outputs.txt
../Lecture20/all_fastas.faa
./
```

no changes added to commit (use "git add" and/or "git commit -a")

List all files (not folders) that contain the text string `nake` (case insensitive)

```
ls -al | grep ^- | grep -i "nake" | awk '{print $NF}'
```

```
foosnake
function_snake
Snakefile1
Snakefile1_named
SnakefileEx1
velvet_snakefile1
velvet_snakefile2
velvet_snakefile3
velvet_snakefile4
```

Put them all into a variable

```
wanted_snakes=$(ls -al | grep ^- | grep -i "nake" | awk '{print $NF}')
```

```
echo ${wanted_snakes}
```

```
foosnake function_snake Snakefile1 Snakefile1_named SnakefileEx1
velvet_snakefile1 velvet_snakefile2 velvet_snakefile3 velvet_snakefile4
```

You might want to add other files?

```
git add ${wanted_snakes}
```

```
git commit -m "Various snakefiles from the snakemake lecture"
```

```
[master c4a2579] Various snakefiles from the snakemake lecture
 9 files changed, 149 insertions(+)
 create mode 100755 Lecture22/Snakefile1
 create mode 100755 Lecture22/Snakefile1_named
 create mode 100755 Lecture22/SnakefileEx1
 create mode 100755 Lecture22/foosnake
 create mode 100755 Lecture22/function_snake
 create mode 100755 Lecture22/velvet_snakefile1
 create mode 100755 Lecture22/velvet_snakefile2
 create mode 100755 Lecture22/velvet_snakefile3
 create mode 100755 Lecture22/velvet_snakefile4
```

Push our lectures repository to the one we just created at GitHub

```
git push AllLectures master
```

```
Enumerating objects: 17, done.
Counting objects: 100% (17/17), done.
Delta compression using up to 256 threads
Compressing objects: 100% (15/15), done.
Writing objects: 100% (16/16), 3.69 KiB | 68.00 KiB/s, done.
Total 16 (delta 7), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (7/7), completed with 1 local object.
To https://github.com/B123456-2121/BPSM_LectureStuff.git
 75d3a16..c4a2579  master -> master
```

Finally....

git log

```
c4a2579 (HEAD -> master, AllLectures/master) Various snakefiles from the snakemake lecture
bf575a2 Scripts from the BioPython lecture
75d3a16 Scripts from the python plots lecture
41e1ac8 Scripts from the python regex lecture
097800e Script from the python pandas lecture
1d2319e Scripts from the python dictionaries lecture
c49df45 Scripts from the python functions lecture
3b8bf41 My scripts from the python conditionals exercises Py04
8f58f4c Scripts from the python lists and loops lecture 3
e444ef8 Scripts from the second BPSM python lecture
6d6f5e7 My python scripts from the first BPSM python lecture
eae65c6 (Lecture07/master) edirect seqs related stuff added
eea4f5e (Lec_6_branch) Kept bash script output files from Lecture 06
fbc5cc0 Added run_this_script_v1 from BPSM Lecture06
ce2c15a nematode BLASTDB stuff added
19cdd2d Added the example_people_data.tsv file from Lecture03
137c3db Decided to keep blastoutput2.out
08d379b Decided to keep myinputfile.tsv
c9b5b18 First attempt at parsing BLAST data
bbf825c My first awk script, Lecture05
68f27fb There were conflicts with someotherfile.sh, kept all changes
83629cb (Version1.2) Final version of the someotherfile.sh shell script prior to merge
0b5425b (tag: v2.0) Updated contents of tar, coolscript
5d8aeaa I am not sure cool script is actually any good
bbd36e7 Added my_cool_script
ea60357 Third version of the someotherfile.sh shell script
b6ae4a7 Updated version of tar
ce230e1 (tag: v1.2) Second version of the someotherfile.sh shell script
a5add7c Added the someotherfile.sh shell script
101b700 Additional motif files, and updated tar file
d21cb3d Adding initial tar file
59d1756 CodeFiles all done and tar zipped
6dd85a8 First file added
```

All good, well done!

Have your say...

The "Postbox of Truth": is your chance to give anonymous "in course" feedback/comments. Your comments are completely anonymous and confidential, so say what you think, irrespective of whether the comments are positive or otherwise! You can use this multiple times too: I use it throughout the course to enable you to give context-relevant feedback/make suggestions.

Just click the image!



[Sources used](#)